

A Quantum Resource Theory of Transpose-Reciprocity Breaking from Non-Hermitian Propagators, with an Exact Witness on the Hatano–Nelson Skin-Effect Orbit

Ann Author

Authors' institution and/or address

(Dated: September 28, 2026)

This work develops a quantum resource theory of channel transpose-reciprocity breaking, motivated by conventional non-Hermitian skin-effect (NHSE) physics rather than by NHSE itself as a universal resource. The primitive fixed-time objects are CPTP bistochastic channels obtained from normalized non-Hermitian propagators through a canonical Julia unitary completion. The Julia completion is a representation convention of the present QRT, not a uniqueness claim about physical open-system realizations. At finite system size, the present object definition is intentionally a compatibility structure: because the locality constants may depend on L , the fixed- L object class coincides with the CPTP bistochastic class; nontrivial locality enters only through L -uniform thermodynamic family conditions. The free objects are transpose-reciprocal channels, while the free transformations are the physically realizable superchannels that preserve the object class and commute with channel transpose on that class. We interpret this covariance requirement as the axiomatic resource-neutrality criterion of the present QRT, not as a complete microscopic model of fabrication or device cost. The theory has two faithful monotones, the channel asymmetry $A_\diamond$ and generalized robustness R_{rob} , with $A_\diamond = 2p_{\text{succ}}^{\text{opt}} - 1$ for one-shot discrimination of a channel from its reciprocal transpose.

The exactly solvable calibration sector is the uniform Hatano–Nelson gauge orbit. At a fixed two-site calibration time, $A_\diamond$ has a closed form and is strictly increasing in $|g|$ under a common strict normalization. Hence $|g|$ and $A_\diamond$ induce the same ordering on that orbit, yielding a rigorous necessary condition for every deterministic free conversion. We do not claim the converse deterministic attenuation law for the full ancilla-assisted superchannel class. In the explicitly defined ancilla-free unitary-congruence subclass, a phase-invariant cosquare-trace scalar makes $|g|$ rigid.

Normalization is treated as an operational gauge. The fixed-time resource support, $A_\diamond = 0$ versus $A_\diamond > 0$, is independent of finite strict scalar normalization, while quantitative values are gauge dependent and uniformly comparable inside a common strict-contraction ball. A per-Hamiltonian tight normalization gives an exactly energy-rescaling-invariant trajectory-derived quantity $A_\diamond^{\text{up}}$. An L -uniform short-time estimate yields a non-vacuous thermodynamic lower bound for the uniform Hatano–Nelson family.

For genuinely coupled chains, the paper does not claim a general many-body reduced-channel theorem. Instead, the buffered analysis concerns the separately defined compressed-contraction Julia channel $\Lambda_V = \tilde{\mathcal{E}}_{P_V K P_V}$, whose buffer convergence follows from propagator locality. The covariance-matrix offset arising in a single-particle Gaussian representation is treated only as a diagnostic and is not identified with part of that channel.

The exact scope of the NHSE correspondence is class dependent. Positivity of the trajectory-derived resource is universally equivalent, under the stated assumptions, to transpose-reciprocity breaking. An exact biconditional with conventional NHSE is obtained only on model classes for which conventional NHSE is independently equivalent to transpose-reciprocity breaking. The uniform Hatano–Nelson family provides an explicit exact example. Reciprocal non-Abelian $\mathbb{Z}_2$ skin physics and genuinely Lindbladian skin topology are outside the universal scope of this channel QRT. The principal open problems are quantitative gauge dependence, deterministic attenuation on the full free-superchannel class, and the relation between compressed single-particle Julia channels and genuine many-body reduced channels.

I. OPERATIONAL STARTING POINT: FIXED-TIME NHSE CHANNELS AS RESOURCES

A quantum resource theory is specified by an object class, a free subset, admissible free transformations, and the induced conversion preorder; resource monotones are nonincreasing under the free transformations [1, 2]. In channel resource theories, the primitive objects are quantum channels and the transformations can be represented by quantum superchannels [2, 3].

Our aim is to formulate a QRT in which transpose-reciprocity breaking is the primitive resource, with conventional NHSE arising as a physically motivated instance

rather than as the object being axiomatized. The primitive resource object is a CPTP bistochastic channel at a fixed operational time t . A time-dependent non-Hermitian evolution gives a family of such objects, but the family is a derived trajectory rather than a new primitive process tensor or comb object.

Remark 1 (Physical status of the reciprocity-covariant free-operation choice). *We adopt covariance under channel transpose as the defining symmetry constraint on the free superchannel class. The motivation is that, in the fixed physical reciprocity frame, transpose exchanges the roles of forward and backward couplings, while transpose-reciprocal Hamiltonians describe devices with symmet-*

ric couplings in that frame, including reciprocal circuit elements, symmetric optical couplers, and Hermitian-coupling lattice realizations with on-site gain or loss.

The present paper should nevertheless distinguish two statements. First, covariance is an axiomatic resource-theoretic choice: it guarantees that free superchannels cannot create transpose-reciprocity breaking from a reciprocal channel and yields the monotonicity theorems below. Second, covariance alone is not a microscopic model of fabrication cost. In particular, the construction

$$\Theta_\Gamma(\mathcal{N}) = \Gamma^{\text{rev}} \circ \mathcal{N} \circ \Gamma \quad (1)$$

of Proposition 12 allows an arbitrary bistochastic Γ from the stated finite-size object class as a processing primitive. We therefore do not interpret the present $\mathfrak{D}_L$ class as a complete statement that every such implementation is physically resource-free in an engineering sense. Rather, it is the symmetry-covariant free-operation class of the QRT constructed here. A stricter hardware-level resource theory would require an additional, independently specified class of resource-neutral pre- and post-processing devices.

A non-Hermitian Hamiltonian by itself does not define a CPTP map. We therefore normalize its propagator to a strict contraction and embed it into a canonical Julia unitary dilation. This completion is an operational representation of the normalized propagator; it is not a claim that every open-system realization of an NHSE is equivalent to a postselected non-Hermitian Hamiltonian, nor that the Julia dilation is the unique physical realization of the contraction.

Definition 1 (Single-particle Hilbert space). *For a chain of length L with internal dimension M , let*

$$\mathcal{H}_L = \text{span} \{|n, \alpha\rangle : 1 \leq n \leq L, 1 \leq \alpha \leq M\}. \quad (2)$$

The spatial basis is fixed once and for all. All reciprocity statements below refer to this basis. When the canonical Julia dilation is formed, the environmental copy is equipped with the corresponding copied spatial basis.

Throughout, $\|\cdot\|_\infty$ denotes the operator (spectral) norm.

Definition 2 (Canonical dilation lattice). *For the canonical Julia completion define*

$$\tilde{\mathcal{H}}_L = \mathcal{H}_L \oplus \mathcal{H}_L^{\text{env}}, \quad \mathcal{H}_L^{\text{env}} \simeq \mathcal{H}_L, \quad (3)$$

and let

$$\tilde{P}_n = P_n \oplus P_n^{\text{env}} \quad (4)$$

be the spatial projector onto the physical and environmental degrees of freedom associated with cell n . The fixed-time channel QRT below acts on $\tilde{\mathcal{H}}_L$.

Remark 2 (Reciprocity frame as operational data). *The transpose operation, and hence the distinction between*

reciprocal and transpose-reciprocity breaking channels, is defined relative to the fixed physical basis specified above. In particular, the resource is a property of the pair consisting of the channel and the chosen reciprocity frame, not of an abstract Hilbert-space operator under arbitrary unitary changes of basis. A basis change is regarded as a mere relabeling only when it preserves the physical reciprocity datum used to define transpose. This qualification is essential for internal degrees of freedom, where a local unitary change of basis can alter the matrix condition $H^T = H$ without changing the underlying laboratory device.

Definition 3 (Admissible normalized propagator). *Let H be a finite-dimensional non-Hermitian Hamiltonian and*

$$U_H(t) = e^{-iHt}. \quad (5)$$

Let $\mathcal{H}_{\text{adm}}$ be a Hamiltonian class closed under transpose, $H \mapsto H^T$, and define

$$\kappa_*(t) := \sup_{H \in \mathcal{H}_{\text{adm}}} \|U_H(t)\|_\infty^2 < \infty \quad (6)$$

for every operational time t . Since

$$\|U_{H^T}(t)\|_\infty = \|U_H(t)\|_\infty, \quad (7)$$

this normalization envelope is itself transpose invariant. Choose a strictly positive function $\varepsilon(t) > 0$ and define the common admissible normalization

$$\kappa_\varepsilon(t) = [1 + \varepsilon(t)] \kappa_*(t). \quad (8)$$

Then

$$K_{H,t} = \frac{U_H(t)}{\sqrt{\kappa_\varepsilon(t)}} \quad (9)$$

is a strict contraction:

$$K_{H,t}^\dagger K_{H,t} \leq \frac{1}{1 + \varepsilon(t)} \mathbb{1}. \quad (10)$$

When short-time normalization information is needed, we assume $\varepsilon(t) \rightarrow 0$ as $t \rightarrow 0^+$. If $\|H\|_\infty \leq h_{\text{max}}$ throughout the admissible class, then

$$\kappa_*(t) \leq e^{2h_{\text{max}}t}, \quad \kappa_\varepsilon(t) \leq [1 + \varepsilon(t)]e^{2h_{\text{max}}t}. \quad (11)$$

Moreover,

$$K_{H^T,t} = K_{H,t}^T. \quad (12)$$

The success branch associated with the normalized propagator is

$$\mathcal{E}_{H,t}^{\text{succ}}(\rho) = K_{H,t} \rho K_{H,t}^\dagger. \quad (13)$$

Proposition 1 (Reciprocity-compatible Julia dilation). *For every contraction K on $\mathcal{H}_L$, define*

$$D_K = (\mathbb{1} - K^\dagger K)^{1/2}, \quad D_{K^\dagger} = (\mathbb{1} - K K^\dagger)^{1/2}, \quad (14)$$

and let

$$\tilde{\mathcal{H}}_L = \mathcal{H}_L \oplus \mathcal{H}_L^{\text{env}}. \quad (15)$$

The Julia operator

$$\tilde{U}_K = \begin{pmatrix} K & D_{K^\dagger} \\ D_K & -K^\dagger \end{pmatrix} \quad (16)$$

is unitary. If $K^T = K$, then

$$\tilde{U}_K^T = \tilde{U}_K. \quad (17)$$

Consequently

$$\tilde{\mathcal{E}}_K(X) = \tilde{U}_K X \tilde{U}_K^\dagger \quad (18)$$

is CPTP, unital, and reciprocal whenever $K^T = K$.

Proof. The Julia operator is unitary for every contraction. The standard defect identities

$$D_{K^\dagger} K = K D_K, \quad D_K K^\dagger = K D_{K^\dagger} \quad (19)$$

together with

$$D_K^2 = \mathbb{1} - K^\dagger K, \quad D_{K^\dagger}^2 = \mathbb{1} - K K^\dagger \quad (20)$$

give

$$\tilde{U}_K^\dagger \tilde{U}_K = \tilde{U}_K \tilde{U}_K^\dagger = \mathbb{1}. \quad (21)$$

If $K^T = K$, then

$$D_K^T = D_{K^\dagger}, \quad D_{K^\dagger}^T = D_K, \quad (22)$$

by functional calculus and $K^* = K^\dagger$ in this case. Hence the Julia block matrix is symmetric. $\square$

Corollary 1 (General transpose covariance of the Julia dilation). *For every contraction K ,*

$$\tilde{U}_K^T = \tilde{U}_{K^T}, \quad \tilde{\mathcal{E}}_K^{\text{rev}} = \tilde{\mathcal{E}}_{K^T}. \quad (23)$$

Hence, for the transpose-closed scalar normalization of Definition 3,

$$\mathcal{N}_{H,t}^{\text{rev}} = \mathcal{N}_{H^T,t}. \quad (24)$$

Proof. For a general matrix K , functional calculus gives the exact identities

$$D_K^T = D_{K^*}, \quad D_{K^\dagger}^T = D_{K^T}. \quad (25)$$

Therefore

$$\tilde{U}_K^T = \begin{pmatrix} K^T & D_{K^*} \\ D_{K^T} & -K^* \end{pmatrix} = \tilde{U}_{K^T}, \quad (26)$$

which proves the first identity. For a unitary channel, channel transpose is generated by the transposed implementing unitary, proving the second. The final statement follows from $K_{H^T,t} = K_{H,t}^T$. $\square$

Remark 3 (Canonical representation choice). *The map*

$$K \mapsto \tilde{U}_K \mapsto \tilde{\mathcal{E}}_K \quad (27)$$

is fixed as the canonical representation convention of the present QRT. The results below therefore concern the chosen Julia-completed channel representative. They do not assert that all Stinespring dilations of the same contraction, or all physical open-system realizations of the same non-Hermitian evolution, are resource-equivalent.

Lemma 1 (Quasi-locality of the Julia dilation). *Suppose H is uniformly bounded and finite range in one dimension, and the normalization is chosen so that $\|K_{H,t}\|_\infty \leq q_t < 1$ uniformly in system size. Then, at each fixed finite time, $U_H(t)$, $K_{H,t}$, $D_{K_{H,t}}$, $D_{K_{H,t}^\dagger}$, and $\tilde{U}_{K_{H,t}}$ are exponentially quasi-local. In particular, there exist $C_t, \mu_t > 0$, independent of L , such that*

$$\|\tilde{P}_n A \tilde{P}_m\| \leq C_t e^{-\mu_t |n-m|} \quad (28)$$

for each operator A in the list.

Proof. If H has range R , then H^r has range at most rR , and

$$P_n U_H(t) P_m = \sum_{r \geq |n-m|/R} \frac{(-it)^r}{r!} P_n H^r P_m. \quad (29)$$

The resulting Poisson tail is exponentially small at fixed t . The same is true of $K_{H,t}$ and $K_{H,t}^\dagger K_{H,t}$. Since $\|K_{H,t}^\dagger K_{H,t}\|_\infty \leq q_t^2 < 1$, the binomial series

$$(\mathbb{1} - X)^{1/2} = \sum_{r=0}^{\infty} c_r X^r, \quad X = K_{H,t}^\dagger K_{H,t}, \quad (30)$$

converges absolutely in operator norm and preserves exponential quasi-locality under the convolution of exponentially decaying kernels. The same applies to $D_{K^\dagger}$, and the block form of $\tilde{U}_K$ completes the proof. $\square$

Remark 4. *Quasi-locality, rather than strict finite range, is the natural class because the defect operators are generally not finite range even when K is.*

Definition 4 (Fixed-time resource channel). *For Hamiltonian H and fixed operational time t , define the primitive resource object*

$$\mathcal{N}_{H,t} := \tilde{\mathcal{E}}_{K_{H,t}}. \quad (31)$$

This definition uses the canonical Julia-completed representative fixed above. The resource is therefore defined relative to the chosen normalization convention and canonical representation choice.

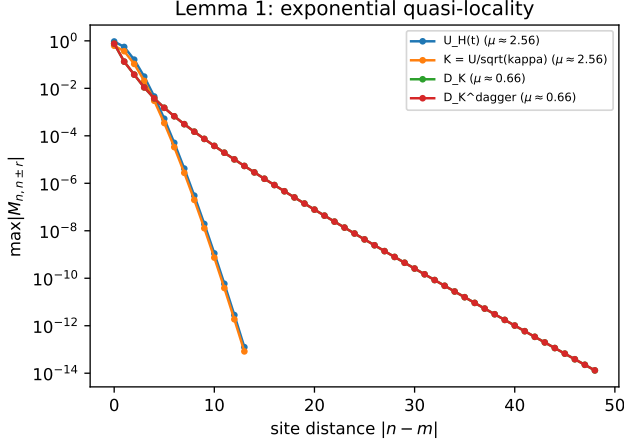


FIG. 1. Numerical consistency check of Lemma 1 for a finite-range, uniformly bounded Hamiltonian at fixed L and fixed operational time t . The propagator $U_H(t)$ and the normalized contraction $K = U_H(t)/\sqrt{\kappa}$ give the same fitted rate ($\mu \approx 2.56$), as expected from the scalar relation between them. For the defect operators, the fit on $r \in [2, 25]$ gives $\mu \approx 0.66$, while a fit over the full available range $r \in [2, 48]$ gives approximately $\mu \approx 0.59$. The fit-window dependence is therefore non-negligible, and no precision exponent is claimed. The robust numerical observation is qualitative exponential decay of the defect operators over the tested range, with visibly slower decay and greater fit sensitivity than K . The $U_H(t)/K$ curves are fit by least squares on $r \in [2, 13]$ before reaching double-precision noise.

Definition 5 (Trajectory-derived channel resource). For a time window $\mathcal{T}$, define the derived trajectory

$$\mathfrak{N}_H : \mathcal{T} \rightarrow \text{Obj}, \quad t \mapsto \mathcal{N}_{H,t}. \quad (32)$$

This map is not treated as a primitive process tensor or quantum comb.

Lemma 2 (Conditional spatial transition probabilities). For a localized physical input $|m\rangle$ and conditioning on successful return to the physical sector,

$$P_t(n|m) = \frac{|U_H(t)_{nm}|^2}{\sum_r |U_H(t)_{rm}|^2}. \quad (33)$$

Proof. The physical success probability at site n is proportional to $|K_{H,t}(n, m)|^2$. The scalar normalization cancels upon conditioning. $\square$

II. RECIPROCITY, LOCALITY, FREE CHANNELS, AND FREE SUPERCHANNELS

Definition 6 (Finite-size exponential compatibility class). For fixed L , let $\mathfrak{Q}_{\text{exp},L}$ be the set of channels $\mathcal{N}_L$ on $\tilde{\mathcal{H}}_L$ for which there exist constants $C, \mu > 0$ and a Kraus

representation

$$\mathcal{N}_L(X) = \sum_a K_{a,L} X K_{a,L}^\dagger \quad (34)$$

satisfying

$$\sum_a \left\| \tilde{P}_n K_{a,L} \tilde{P}_m \right\|^2 \leq C^2 e^{-2\mu|n-m|} \quad (35)$$

for all sites n, m . At fixed L , the constants may depend on the channel and on L .

Because the Hilbert space is finite dimensional, every finite-dimensional channel belongs to $\mathfrak{Q}_{\text{exp},L}$. Indeed, for any fixed $\mu > 0$ and any finite Kraus representation one may choose

$$C = \max_{n,m} e^{\mu|n-m|} \left(\sum_a \left\| \tilde{P}_n K_{a,L} \tilde{P}_m \right\|^2 \right)^{1/2}, \quad (36)$$

which is finite. Thus the fixed- L condition is retained only as a compatibility notation for the thermodynamic locality structure introduced below.

Definition 7 (Bistochastic channel class). For finite L ,

$$\text{Chan}_L^{\text{bi}} = \left\{ \mathcal{N} \in \text{Chan}(\tilde{\mathcal{H}}_L) : \mathcal{N}(\mathbb{1}) = \mathbb{1} \right\}. \quad (37)$$

Because all maps in $\text{Chan}_L^{\text{bi}}$ are CPTP by definition, this is precisely the finite-dimensional class of CPTP bistochastic channels.

Definition 8 (Finite-size object class). For each finite L , define

$$\text{Obj}_L = \text{Chan}_L^{\text{bi}}. \quad (38)$$

Equivalently, this is the diamond-norm closure of $\mathfrak{Q}_{\text{exp},L} \cap \text{Chan}_L^{\text{bi}}$, since $\mathfrak{Q}_{\text{exp},L}$ already contains all finite-dimensional channels at fixed L . Thermodynamic statements refer to sequences drawn from these fixed-size classes with additional L -uniform locality bounds.

Remark 5 (Where locality becomes nontrivial). At fixed finite L , the primitive QRT is therefore a symmetry/covariance resource theory on the bistochastic channel class; the exponential locality condition is not a substantive restriction. All nontrivial locality content enters through the L -uniform family definitions and locality certificates used in the thermodynamic layer. This separation is deliberate and will be used throughout.

Definition 9 (Thermodynamic object family). A sequence $\{\mathcal{N}_L\}_{L \geq 1}$ is a thermodynamically admissible object family if, for every compact operational time interval $\mathcal{T}_0$, there are constants $C_{\mathcal{T}_0}, \mu_{\mathcal{T}_0} > 0$ independent of L and $t \in \mathcal{T}_0$ and Kraus representations of $\mathcal{N}_{L,t}$ such that

$$\sum_a \left\| \tilde{P}_n K_{a,L,t} \tilde{P}_m \right\|^2 \leq C_{\mathcal{T}_0}^2 e^{-2\mu_{\mathcal{T}_0}|n-m|} \quad (39)$$

for all n, m , while $\mathcal{N}_{L,t} \in \text{Obj}_L$ for every L, t .

Definition 10 (Channel transpose). For $\mathcal{N} \in \text{Chan}_L^{\text{bi}}$ with Kraus operators $\{K_a\}$ in the fixed spatial and environmental basis, define

$$\mathcal{N}^{\text{rev}}(X) = \sum_a K_a^T X K_a^*. \quad (40)$$

Lemma 3 (Basic properties of channel transpose). For every $\mathcal{N} \in \text{Chan}_L^{\text{bi}}$:

1. $\mathcal{N}^{\text{rev}} \in \text{Chan}_L^{\text{bi}}$;
2. $(\mathcal{N}^{\text{rev}})^{\text{rev}} = \mathcal{N}$;
3. if $\mathcal{U}(X) = UXU^\dagger$, then $\mathcal{U}^{\text{rev}}(X) = U^T X U^*$;
4. the definition is independent of the chosen Kraus representation;
5. transpose preserves Obj_L ;
6. the map $\mathcal{N} \mapsto \mathcal{N}^{\text{rev}}$ is linear and continuous in finite dimension.

Proof. Set $L_a = K_a^T$. Complete positivity is immediate. Trace preservation follows from bistochasticity:

$$\sum_a L_a^\dagger L_a = \sum_a K_a^* K_a^T = \left(\sum_a K_a K_a^\dagger \right)^T = \mathbb{1}. \quad (41)$$

Unitality follows from the original trace-preserving condition:

$$\mathcal{N}^{\text{rev}}(\mathbb{1}) = \sum_a K_a^T K_a^* = \left(\sum_a K_a^\dagger K_a \right)^T = \mathbb{1}. \quad (42)$$

Double transpose returns the original Kraus operators. If $L_b = \sum_a u_{ba} K_a$ is another Kraus representation, then $L_b^T = \sum_a u_{ba} K_a^T$ and

$$\sum_b L_b^T X L_b^* = \sum_{a,a'} \left(\sum_b u_{ba} u_{ba'}^* \right) K_a^T X K_{a'}^* = \sum_a K_a^T X K_a^*, \quad (43)$$

so the channel is representation independent.

Finally,

$$\tilde{P}_n K_a^T \tilde{P}_m = (\tilde{P}_m K_a \tilde{P}_n)^T, \quad (44)$$

so the same locality bound holds after exchanging the two sites; diamond-norm closure is preserved by continuity. Linearity and continuity are immediate in finite dimension because rev is a fixed linear involution on the finite-dimensional vector space of superoperators. $\square$

Remark 6 (Transpose is not assumed to be a diamond isometry). The channel-transpose involution is not used anywhere below as a diamond-norm isometry. In particular, in general one cannot assert

$$\|\mathcal{A}^{\text{rev}} - \mathcal{B}^{\text{rev}}\|_\diamond = \|\mathcal{A} - \mathcal{B}\|_\diamond. \quad (45)$$

Whenever continuity of rev is needed, it is stated either as a separate finite-dimensional continuity bound or directly as an assumption on both a channel and its transpose.

Lemma 4 (Convexity and anti-homomorphism). For $\mathcal{A}, \mathcal{B} \in \text{Chan}_L^{\text{bi}}$ and $\alpha + \beta = 1$,

$$(\alpha\mathcal{A} + \beta\mathcal{B})^{\text{rev}} = \alpha\mathcal{A}^{\text{rev}} + \beta\mathcal{B}^{\text{rev}}, \quad (46)$$

while

$$(\mathcal{A} \circ \mathcal{B})^{\text{rev}} = \mathcal{B}^{\text{rev}} \circ \mathcal{A}^{\text{rev}}. \quad (47)$$

Also,

$$(\mathcal{N} \otimes \mathbb{1}_E)^{\text{rev}} = \mathcal{N}^{\text{rev}} \otimes \mathbb{1}_E, \quad \left(\bigotimes_k \mathcal{N}_k \right)^{\text{rev}} = \bigotimes_k \mathcal{N}_k^{\text{rev}}. \quad (48)$$

Proof. For composition, the Kraus operators are $A_a B_b$, and

$$(A_a B_b)^T = B_b^T A_a^T. \quad (49)$$

The tensor identities follow from tensor-product Kraus representations. $\square$

Remark 7 (Why covariance is imposed on the complete superchannel). Because channel transpose reverses composition order, a reciprocal pre-processing map and a reciprocal post-processing map do not by themselves imply covariance. The correct free-operation condition is therefore

$$\Theta(\mathcal{N}^{\text{rev}}) = [\Theta(\mathcal{N})]^{\text{rev}} \quad (50)$$

for the complete superchannel. This is stronger and better typed than imposing a separate reciprocity condition on the individual pre- and post-processing components.

Remark 8 (Covariance is a genuine restriction, not a vacuous one). It is worth exhibiting explicitly that the composite covariance condition rules out superchannels built from individually non-reciprocal, and otherwise unremarkable, processing maps. Take $\Gamma_{\text{pre}} = \Gamma_{\text{post}} = \mathcal{V}$ for a fixed unitary conjugation $\mathcal{V}(X) = VXV^\dagger$ with V an arbitrary (non-symmetric) unitary, and no ancilla. Then

$$\Theta(\mathcal{N}) = \mathcal{V} \circ \mathcal{N} \circ \mathcal{V}, \quad [\Theta(\mathcal{N})]^{\text{rev}} = \mathcal{V}^{\text{rev}} \circ \mathcal{N}^{\text{rev}} \circ \mathcal{V}^{\text{rev}} \quad (51)$$

by Lemma 4, while

$$\Theta(\mathcal{N}^{\text{rev}}) = \mathcal{V} \circ \mathcal{N}^{\text{rev}} \circ \mathcal{V}. \quad (52)$$

Covariance for all $\mathcal{N}$ requires $\mathcal{V} \circ \mathcal{M} \circ \mathcal{V} = \mathcal{V}^{\text{rev}} \circ \mathcal{M} \circ \mathcal{V}^{\text{rev}}$ for all $\mathcal{M} = \mathcal{N}^{\text{rev}}$ in the object class. Taking $\mathcal{M}$ to be unitary channels with implementing unitary W ranging over a sufficiently rich set (e.g. all local unitaries), this requires $VWV \propto V^T W V^T$ for all such W , and hence $A := (V^T)^{-1} V \propto \mathbb{1}$, i.e. $V^T \propto V$. Thus for V with $V^T \not\propto V$ the superchannel $\mathcal{N} \mapsto \mathcal{V} \circ \mathcal{N} \circ \mathcal{V}$ is not free, even though $\mathcal{V}$ is an innocuous unitary; conversely, if $V^T \propto V$ then $\mathcal{V}^{\text{rev}} = \mathcal{V}$ and covariance holds. The paired choice $\Gamma_{\text{post}} = \Gamma_{\text{pre}}^{\text{rev}}$ of Proposition 12 restores covariance for arbitrary Γ_{pre} . The covariance constraint is therefore a nontrivial selection rule on how pre- and post-processing must be correlated, not a condition satisfied by generic physical processing.

Definition 11 (Free channel set). For each finite L , define

$$\mathfrak{F}_L = \{\mathcal{N} \in \text{Obj}_L : \mathcal{N}^{\text{rev}} = \mathcal{N}\}. \quad (53)$$

Thus the free channels are precisely the reciprocal subset of the allowed quasi-local bistochastic channels.

Proposition 2 (Hamiltonian reciprocity implies channel reciprocity). If $H^T = H$, then

$$U_H(t)^T = U_H(t), \quad K_{H,t}^T = K_{H,t}, \quad (54)$$

and hence

$$\mathcal{N}_{H,t}^{\text{rev}} = \mathcal{N}_{H,t}. \quad (55)$$

Proof. Transpose commutes with the exponential:

$$U_H(t)^T = e^{-iH^T t} = e^{-iHt} = U_H(t), \quad (56)$$

and the normalization is a transpose-invariant positive scalar. Proposition 1 then gives reciprocity. $\square$

A. Free superchannels and a thermodynamic locality certificate

Definition 12 (Physical finite-size free superchannel). A deterministic quantum superchannel has the physical realization

$$\Theta(\mathcal{N}) = \Gamma_{\text{post}} \circ (\mathcal{N} \otimes \mathbb{1}_E) \circ \Gamma_{\text{pre}}, \quad (57)$$

where $\Gamma_{\text{pre}}, \Gamma_{\text{post}}$ are CPTP maps and E is an auxiliary memory system [3].

A finite-size superchannel is free if:

1. it is physically realizable;
2. it preserves the object class,

$$\Theta(\text{Obj}_L) \subseteq \text{Obj}_L; \quad (58)$$

3. it is reciprocity-covariant,

$$\Theta(\mathcal{N}^{\text{rev}}) = [\Theta(\mathcal{N})]^{\text{rev}} \quad \forall \mathcal{N} \in \text{Obj}_L. \quad (59)$$

The covariance and object-preservation requirements are conditions on the complete superchannel, not on the individual pre- and post-processing maps separately. We denote this finite-size class by $\mathfrak{D}_L$.

Definition 13 (Quasi-local CPTP preprocessing/postprocessing certificate). Let Γ_L be a CPTP map whose input and output lattices carry the fixed spatial basis. We say that Γ_L is (C_Γ, μ_Γ) quasi-local in the Heisenberg sense if, for every finite interval V , every $r \geq 0$, and every observable O_V supported on

V , there exists an observable $O_{V^{(+)r}}$ supported on the r -neighborhood of V such that

$$\left\| \Gamma_L^\dagger(O_V) - O_{V^{(+)r}} \right\|_\infty \leq C_\Gamma |V| e^{-\mu_\Gamma r} \|O_V\|_\infty. \quad (60)$$

A finite-depth bounded-range circuit channel is a special case, with an error that vanishes exactly beyond a finite light cone.

Definition 14 (Locality certificate for a superchannel family). A thermodynamic family $\{\Theta_{L,t}\}_{L \geq 1, t \in \mathcal{T}_0}$ admits a locality certificate on a compact operational time interval $\mathcal{T}_0$ if there are ancillary memories E_L decomposed into spatial cells with uniformly bounded local dimension, and physical realizations

$$\Theta_{L,t}(\mathcal{N}) = \Gamma_{\text{post},L,t} \circ (\mathcal{N} \otimes \mathbb{1}_{E_L}) \circ \Gamma_{\text{pre},L,t}, \quad (61)$$

for which both $\Gamma_{\text{pre},L,t}$ and $\Gamma_{\text{post},L,t}$ satisfy the Heisenberg quasi-locality condition in Definition 13 with constants bounded independently of L and $t \in \mathcal{T}_0$. This certificate is required only at the family level; no covariance is imposed on either component separately.

A time-independent superchannel family is included as the special case $\Theta_{L,t} = \Theta_L$.

Definition 15 (Thermodynamic free-superchannel family). A collection

$$\{\Theta_{L,t}\}_{L \geq 1, t \in \mathcal{T}_0} \quad (62)$$

is a thermodynamically free superchannel family on $\mathcal{T}_0$ if each $\Theta_{L,t} \in \mathfrak{D}_L$ and the family admits a locality certificate with constants uniform in L and $t \in \mathcal{T}_0$. We denote the class of such families by $\mathfrak{D}_\infty$.

Remark 9 (Non-vacuous thermodynamic locality). The family-level definition is the actual thermodynamic locality condition used in all thermodynamic monotonicity statements. Thus an arbitrary finite-size global channel is not automatically promoted to a thermodynamically free transformation merely because it happens to belong to $\mathfrak{D}_L$ at every fixed L .

Definition 16 (Fixed-time conversion preorder). For $\mathcal{N}, \mathcal{M} \in \text{Obj}_L$, write

$$\mathcal{N} \succeq_{\mathfrak{D}} \mathcal{M} \quad (63)$$

if there exists a finite-size free superchannel $\Theta \in \mathfrak{D}_L$ such that $\Theta(\mathcal{N}) = \mathcal{M}$.

Proposition 3 (Basic closure properties of the free transformations). The identity superchannel is free, the composition of two free superchannels is free, and every free superchannel preserves the free set:

$$\mathcal{N} \in \mathfrak{F}_L \implies \Theta(\mathcal{N}) \in \mathfrak{F}_L \quad (\Theta \in \mathfrak{D}_L). \quad (64)$$

Consequently $\succeq_{\mathfrak{D}}$ is a preorder on Obj_L .

Proof. The identity superchannel is physical, preserves the object class, and trivially commutes with rev. For $\Theta_1, \Theta_2 \in \mathfrak{D}_L$,

$$(\Theta_2 \circ \Theta_1)(\mathcal{N}^{\text{rev}}) = \Theta_2(\Theta_1(\mathcal{N})^{\text{rev}}) = \Theta_2(\Theta_1(\mathcal{N}))^{\text{rev}}, \quad (65)$$

so covariance is preserved by composition; object preservation and physical realizability compose as well. Finally, if $\mathcal{N} = \mathcal{N}^{\text{rev}}$, covariance gives $\Theta(\mathcal{N}) = \Theta(\mathcal{N}^{\text{rev}}) = \Theta(\mathcal{N})^{\text{rev}}$, hence $\Theta(\mathcal{N}) \in \mathfrak{F}_L$. $\square$

Remark 10 (Formal reciprocity projection). *The affine map*

$$\Pi_{\text{rec}}(\mathcal{N}) = \frac{1}{2}(\mathcal{N} + \mathcal{N}^{\text{rev}}) \quad (66)$$

always produces a reciprocal bistochastic channel. It is a formal proof device and is not assumed to be an operationally free superchannel on an unknown input.

III. FAITHFUL FIXED-TIME CHANNEL RESOURCE MONOTONES

Definition 17 (Channel asymmetry). *Define*

$$A_{\diamond}(\mathcal{N}) = \frac{1}{2}\|\mathcal{N} - \mathcal{N}^{\text{rev}}\|_{\diamond}. \quad (67)$$

Proposition 4 (Operational channel-reciprocity discrimination). *Suppose one receives a black-box channel that is promised to be either $\mathcal{N}$ or $\mathcal{N}^{\text{rev}}$ with equal prior probabilities. Then the optimal one-shot discrimination success probability is*

$$p_{\text{succ}}^{\text{opt}} = \frac{1}{2} + \frac{1}{4}\|\mathcal{N} - \mathcal{N}^{\text{rev}}\|_{\diamond} = \frac{1 + A_{\diamond}(\mathcal{N})}{2}. \quad (68)$$

Equivalently,

$$A_{\diamond}(\mathcal{N}) = 2p_{\text{succ}}^{\text{opt}} - 1 = 2\left(p_{\text{succ}}^{\text{opt}} - \frac{1}{2}\right). \quad (69)$$

Thus $A_{\diamond}$ is exactly twice the optimal one-shot advantage over random guessing for detecting transpose-reciprocity breaking.

Proof. The Helstrom theorem for discrimination of quantum channels gives

$$p_{\text{succ}}^{\text{opt}}(\mathcal{N}, \mathcal{M}) = \frac{1}{2} + \frac{1}{4}\|\mathcal{N} - \mathcal{M}\|_{\diamond}. \quad (70)$$

Set $\mathcal{M} = \mathcal{N}^{\text{rev}}$. $\square$

Remark 11 (Operational meaning). *The diamond norm optimizes over reference systems:*

$$\|\mathcal{N} - \mathcal{N}^{\text{rev}}\|_{\diamond} = \sup_{\rho_{RE}} \|(\mathcal{N} \otimes \mathbb{1}_R)(\rho_{RE}) - (\mathcal{N}^{\text{rev}} \otimes \mathbb{1}_R)(\rho_{RE})\|_1. \quad (71)$$

Thus $A_{\diamond}$ compares the coherent quantum processes themselves rather than only classical transition probabilities.

Theorem 1 (Monotonicity and faithfulness of channel asymmetry). *For every free superchannel $\Theta \in \mathfrak{D}_L$,*

$$A_{\diamond}(\Theta(\mathcal{N})) \leq A_{\diamond}(\mathcal{N}). \quad (72)$$

Moreover,

$$A_{\diamond}(\mathcal{N}) = 0 \iff \mathcal{N} \in \mathfrak{F}_L. \quad (73)$$

Proof. Covariance gives

$$[\Theta(\mathcal{N})]^{\text{rev}} = \Theta(\mathcal{N}^{\text{rev}}), \quad (74)$$

and hence

$$A_{\diamond}(\Theta(\mathcal{N})) = \frac{1}{2}\|\Theta(\mathcal{N}) - \Theta(\mathcal{N}^{\text{rev}})\|_{\diamond}. \quad (75)$$

Any physical superchannel is contractive for the diamond norm, giving the inequality. Faithfulness is immediate from the definition of $\mathfrak{F}_L$. $\square$

Definition 18 (Generalized channel robustness). *Define*

$$R_{\text{rob}}(\mathcal{N}) = \inf \left\{ s \geq 0 : \frac{\mathcal{N} + s\mathcal{M}}{1+s} \in \mathfrak{F}_L, \exists \mathcal{M} \in \text{Obj}_L \right\}. \quad (76)$$

Theorem 2 (Faithfulness and monotonicity of robustness). *For every free superchannel $\Theta \in \mathfrak{D}_L$,*

$$R_{\text{rob}}(\Theta(\mathcal{N})) \leq R_{\text{rob}}(\mathcal{N}), \quad (77)$$

and

$$R_{\text{rob}}(\mathcal{N}) = 0 \iff \mathcal{N} \in \mathfrak{F}_L. \quad (78)$$

Proof. If

$$\mathcal{F} = \frac{\mathcal{N} + s\mathcal{M}}{1+s} \in \mathfrak{F}_L, \quad (79)$$

then covariance and linearity give

$$\Theta(\mathcal{F}) = \frac{\Theta(\mathcal{N}) + s\Theta(\mathcal{M})}{1+s} \in \mathfrak{F}_L. \quad (80)$$

Thus the same s is feasible after Θ .

Conversely, if $R_{\text{rob}}(\mathcal{N}) = 0$, there is a sequence $s_k \rightarrow 0$ and $\mathcal{F}_k \in \mathfrak{F}_L$ with

$$\mathcal{F}_k = \frac{\mathcal{N} + s_k\mathcal{M}_k}{1+s_k}. \quad (81)$$

The finite-dimensional channel set is compact, so the feasible $\mathcal{M}_k$ are uniformly bounded in diamond norm. Hence $\|\mathcal{F}_k - \mathcal{N}\|_{\diamond} \rightarrow 0$; closedness of $\mathfrak{F}_L$ then implies $\mathcal{N} \in \mathfrak{F}_L$. $\square$

Remark 12 (Attainment of the robustness infimum). The infimum defining $R_{\text{rob}}(\mathcal{N})$ is in fact attained for every $\mathcal{N} \in \text{Obj}_L$ at finite L . The feasible set

$$\left\{ (s, \mathcal{M}) \in [0, \infty) \times \text{Obj}_L : \frac{\mathcal{N} + s\mathcal{M}}{1+s} \in \mathfrak{F}_L \right\} \quad (82)$$

is nonempty (Proposition 5 exhibits $s = 1$, $\mathcal{M} = \mathcal{N}^{\text{rev}}$ as feasible), and for s ranging over any compact interval $[0, 1]$ the set of admissible $\mathcal{M} \in \text{Obj}_L$ is compact (closed and bounded in the finite-dimensional diamond norm), with the constraint $\frac{\mathcal{N} + s\mathcal{M}}{1+s} \in \mathfrak{F}_L$ defining a closed subset of this compact set by continuity of $(s, \mathcal{M}) \mapsto \frac{\mathcal{N} + s\mathcal{M}}{1+s}$ and closedness of $\mathfrak{F}_L$. The projection of a closed subset of a compact set onto the s -coordinate is compact, hence closed, so the infimum over s is attained by a minimizing pair $(s^*, \mathcal{M}^*)$. This attainment is used implicitly throughout (e.g. in Proposition 13's construction of a specific feasible $\mathcal{M}$ realizing a target robustness value) and is recorded here explicitly for completeness.

Proposition 5 (Global boundedness and tensor-power collapse). For every finite-size channel,

$$0 \leq A_{\diamond}(\mathcal{N}) \leq R_{\text{rob}}(\mathcal{N}) \leq 1. \quad (83)$$

For every $n \geq 1$,

$$A_{\diamond}(\mathcal{N}^{\otimes n}) \leq 1, \quad R_{\text{rob}}(\mathcal{N}^{\otimes n}) \leq 1, \quad (84)$$

so

$$\limsup_{n \rightarrow \infty} \frac{A_{\diamond}(\mathcal{N}^{\otimes n})}{n} = 0, \quad \limsup_{n \rightarrow \infty} \frac{R_{\text{rob}}(\mathcal{N}^{\otimes n})}{n} = 0. \quad (85)$$

Thus the global fixed-time channel measures in this normalization are persistence-type quantities rather than extensive tensor-power rates.

Proof. The diamond distance between two channels is at most 2, hence $A_{\diamond} \leq 1$. For robustness, choose $\mathcal{M} = \mathcal{N}^{\text{rev}}$:

$$\frac{\mathcal{N} + \mathcal{N}^{\text{rev}}}{2} \in \mathfrak{F}_L. \quad (86)$$

This gives $R_{\text{rob}} \leq 1$. The same argument applies to tensor powers using

$$(\mathcal{N}^{\otimes n})^{\text{rev}} = (\mathcal{N}^{\text{rev}})^{\otimes n}. \quad (87)$$

□

Corollary 2 (Robustness dominates channel asymmetry). For every channel,

$$A_{\diamond}(\mathcal{N}) \leq R_{\text{rob}}(\mathcal{N}). \quad (88)$$

Proof. If $s, \mathcal{M}$ are feasible, then the free mixture obeys

$$\mathcal{N} - \mathcal{N}^{\text{rev}} = -s(\mathcal{M} - \mathcal{M}^{\text{rev}}), \quad (89)$$

and the diamond distance between two channels is at most 2. Hence $A_{\diamond}(\mathcal{N}) \leq s$, and infimization gives the result. □

A. Thermodynamic persistence

Definition 19 (Thermodynamic channel family). A fixed-time family $\{\mathcal{N}_{L,t}\}_{L \geq 1}$ is thermodynamically admissible if, for every compact operational time interval $\mathcal{T}_0$, the family admits quasi-local Kraus representations with constants $C_{\mathcal{T}_0}, \mu_{\mathcal{T}_0} > 0$ independent of L and $t \in \mathcal{T}_0$, and each $\mathcal{N}_{L,t}$ belongs to Obj_L .

Definition 20 (Thermodynamic persistence). Define

$$A_{\diamond,\infty}(t) = \limsup_{L \rightarrow \infty} A_{\diamond}(\mathcal{N}_{L,t}), \quad (90)$$

and

$$R_{\text{rob},\infty}(t) = \limsup_{L \rightarrow \infty} R_{\text{rob}}(\mathcal{N}_{L,t}). \quad (91)$$

These are persistence quantities, not extensive densities.

Proposition 6 (Thermodynamic monotonicity). For a thermodynamically free superchannel family $\{\Theta_L\}_{L \geq 1} \in \mathfrak{D}_{\infty}$,

$$A_{\diamond,\infty}[\Theta(\mathcal{N})] \leq A_{\diamond,\infty}(\mathcal{N}), \quad R_{\text{rob},\infty}[\Theta(\mathcal{N})] \leq R_{\text{rob},\infty}(\mathcal{N}). \quad (92)$$

Proof. Apply the finite-size monotonicity inequalities at every L and take the thermodynamic limsup. The family-level locality certificate is what makes the resulting sequence an element of the thermodynamic free-operation class rather than merely a list of unrelated finite-size maps. □

IV. NORMALIZATION GAUGE AS PART OF THE OPERATIONAL SPECIFICATION

The channel resource defined in this work depends on a scalar normalization of the non-Hermitian propagator. This dependence is not an ambiguity to be hidden: the normalization convention is part of the operational specification of the present channel QRT. What is intrinsic within the strict scalar class is the reciprocity support $A_{\diamond} = 0$ versus $A_{\diamond} > 0$, whereas the numerical value of $A_{\diamond}$ depends quantitatively on the chosen normalization gauge.

Definition 21 (Admissible scalar normalization gauge). Let $\mathcal{H}_{\text{adm}}$ be a Hamiltonian class closed under transpose, $H \mapsto H^T$. An admissible scalar normalization gauge is a positive function

$$\kappa : \mathcal{H}_{\text{adm}} \times \mathcal{T} \rightarrow (0, \infty), \quad (H, t) \mapsto \kappa(H, t), \quad (93)$$

satisfying the following conditions.

1. Strict contraction:

$$\kappa(H, t) > \|U_H(t)\|_{\infty}^2, \quad U_H(t) = e^{-iHt}. \quad (94)$$

2. **Transpose invariance:**

$$\boxed{\kappa(H^T, t) = \kappa(H, t).} \quad (95)$$

3. **Thermodynamic uniformity:** for every thermodynamic family $\{H_L\}_{L \geq 1}$ and every compact interval $\mathcal{T}_0 \ni 0$ on which

$$\sup_L \sup_{t \in \mathcal{T}_0} \|U_{H_L}(t)\|_\infty < \infty, \quad (96)$$

there are finite constants $0 < \kappa_- < \kappa_+ < \infty$ such that

$$\boxed{\kappa_- \leq \kappa(H_L, t) \leq \kappa_+} \quad (97)$$

uniformly in L and $t \in \mathcal{T}_0$.

4. **Short-time regularity:** when a short-time expansion is invoked, there exists $\kappa_0 > 0$ such that

$$\boxed{\kappa(H_L, t) = \kappa_0 + O(t)} \quad (98)$$

uniformly in L within the thermodynamic family under consideration.

Given such a gauge, define

$$\boxed{K_{H,t}^{(\kappa)} = \frac{U_H(t)}{\sqrt{\kappa(H, t)}},} \quad (99)$$

and

$$\boxed{\mathcal{N}_{H,t}^{(\kappa)} = \tilde{\mathcal{E}}_{K_{H,t}^{(\kappa)}}} \quad (100)$$

be its canonical Julia-completed channel.

Remark 13 (Operational status of the normalization gauge). The scalar normalization gauge is part of the operational definition of the QRT. Accordingly, the statement $A_\diamond(\mathcal{N}_{H,t}^{(\kappa)})$ refers to a channel resource evaluated after the gauge κ has been fixed. The present theory does not claim that the numerical magnitude of $A_\diamond$ is a representation-independent invariant of the bare Hamiltonian H under arbitrary changes of normalization. What is proved below is that the reciprocity support of the resource is invariant within the admissible strict scalar class, and that quantitative values are equivalent up to explicit continuity bounds inside a common strict-contraction region.

Proposition 7 (Exact reciprocity criterion for a strict Julia dilation). Let K satisfy

$$\|K\|_\infty \leq q < 1. \quad (101)$$

Then

$$\boxed{\tilde{\mathcal{E}}_K^{\text{rev}} = \tilde{\mathcal{E}}_K \iff K^T = K.} \quad (102)$$

Proof. The forward implication from $K^T = K$ is Proposition 1. Conversely, equality of the two unitary-conjugation channels implies

$$\tilde{U}_K^T = e^{i\phi} \tilde{U}_K \quad (103)$$

for some phase. Transposing again gives $e^{2i\phi} = 1$. If $\phi = \pi$, the upper-right block would satisfy

$$D_K^T = -D_{K^\dagger}. \quad (104)$$

Taking traces gives

$$\text{Tr}(D_K) = -\text{Tr}(D_{K^\dagger}), \quad (105)$$

which is impossible because both defect operators are positive definite and hence have strictly positive traces. Thus $\phi = 0$, and the $(1, 1)$ block gives $K^T = K$. $\square$

Proposition 8 (Gauge-invariant reciprocity support). Let κ_1 and κ_2 be any two admissible strict scalar normalization gauges. Then for every finite-dimensional H and every operational time t ,

$$\boxed{A_\diamond(\mathcal{N}_{H,t}^{(\kappa_1)}) = 0 \iff A_\diamond(\mathcal{N}_{H,t}^{(\kappa_2)}) = 0.} \quad (106)$$

More precisely,

$$\boxed{A_\diamond(\mathcal{N}_{H,t}^{(\kappa)}) = 0 \iff U_H(t)^T = U_H(t).} \quad (107)$$

Proof. For an admissible scalar normalization,

$$K_{H,t}^{(\kappa)} = \frac{U_H(t)}{\sqrt{\kappa(H, t)}}, \quad (108)$$

and therefore

$$\left(K_{H,t}^{(\kappa)}\right)^T = K_{H,t}^{(\kappa)} \iff U_H(t)^T = U_H(t). \quad (109)$$

Because the contraction is strict, Proposition 7 applies and gives

$$A_\diamond(\mathcal{N}_{H,t}^{(\kappa)}) = 0 \iff \left(K_{H,t}^{(\kappa)}\right)^T = K_{H,t}^{(\kappa)}. \quad (110)$$

Combining the two equivalences proves the result. $\square$

Proposition 9 (Uniform quantitative equivalence of normalization gauges). Let κ_1 and κ_2 be two normalization gauges for the same propagator $U_H(t)$, and suppose that on the domain of interest

$$\|K_j(H, t)\|_\infty \leq q < 1, \quad K_j(H, t) = \frac{U_H(t)}{\sqrt{\kappa_j(H, t)}}, \quad j = 1, 2. \quad (111)$$

Then

$$\boxed{\left|A_\diamond(\mathcal{N}_{H,t}^{(\kappa_1)}) - A_\diamond(\mathcal{N}_{H,t}^{(\kappa_2)})\right| \leq C(q) \|U_H(t)\|_\infty \left|\kappa_1(H, t)^{-1/2} - \kappa_2(H, t)^{-1/2}\right|} \quad (112)$$

where

$$\boxed{C(q) = 2 \left(1 + \frac{q}{\sqrt{1-q^2}}\right).} \quad (113)$$

Proof. Set

$$X_j = K_j^\dagger K_j, \quad D_{K_j} = (\mathbb{1} - X_j)^{1/2}. \quad (114)$$

Then

$$\|X_1 - X_2\|_\infty \leq (\|K_1\|_\infty + \|K_2\|_\infty) \|K_1 - K_2\|_\infty \leq 2q \|K_1 - K_2\|_\infty. \quad (115)$$

Since the spectra of X_j lie in $[0, q^2]$, the square-root functional calculus on $\mathbb{1} - X$ gives

$$\|D_{K_1} - D_{K_2}\|_\infty \leq \frac{q}{\sqrt{1 - q^2}} \|K_1 - K_2\|_\infty. \quad (116)$$

The same estimate holds for the complementary defect $D_{K_j^\dagger}$. Hence

$$\|\tilde{U}_{K_1} - \tilde{U}_{K_2}\|_\infty \leq \left(1 + \frac{q}{\sqrt{1 - q^2}}\right) \|K_1 - K_2\|_\infty. \quad (117)$$

For unitary channels,

$$\|\mathcal{U}_1 - \mathcal{U}_2\|_\diamond \leq 2 \|U_1 - U_2\|_\infty, \quad (118)$$

and the same estimate applies to the transposed unitary channels. Using the triangle inequality in the definition of $A_\diamond$ gives

$$|A_\diamond(\mathcal{N}_1) - A_\diamond(\mathcal{N}_2)| \leq 2 \left(1 + \frac{q}{\sqrt{1 - q^2}}\right) \|K_1 - K_2\|_\infty. \quad (119)$$

Finally,

$$\|K_1 - K_2\|_\infty = \|U_H(t)\|_\infty \left| \kappa_1(H, t)^{-1/2} - \kappa_2(H, t)^{-1/2} \right|, \quad (120)$$

which proves the claim. $\square$

Proposition 10 (Large-normalization degeneration). *For fixed H and t , let*

$$K^{(\kappa)} = \frac{U_H(t)}{\sqrt{\kappa}}, \quad \kappa > \|U_H(t)\|_\infty^2. \quad (121)$$

Then

$$K^{(\kappa)} \longrightarrow 0, \quad D_{K^{(\kappa)}} \longrightarrow \mathbb{1}, \quad D_{K^{(\kappa)}^\dagger} \longrightarrow \mathbb{1}, \quad (122)$$

and therefore

$$\tilde{U}_{K^{(\kappa)}} \longrightarrow \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix} =: S. \quad (123)$$

Since $S^T = S$,

$$\lim_{\kappa \rightarrow \infty} A_\diamond(\tilde{\mathcal{E}}_{K^{(\kappa)}}) = 0. \quad (124)$$

Thus the numerical value of the channel resource cannot be regarded as a normalization-independent quantity under arbitrary scalar rescaling.

Proof. The convergence of $K^{(\kappa)}$ is immediate. Functional calculus gives $D_{K^{(\kappa)}} \rightarrow \mathbb{1}$ and $D_{K^{(\kappa)}^\dagger} \rightarrow \mathbb{1}$, hence the Julia operator converges to S . Continuity of unitary channels in the diamond norm then implies convergence of the corresponding channel asymmetry. $\square$

Theorem 3 (Normalization-gauge principle). *Within the strict scalar normalization class, the present QRT has the following separation between intrinsic and gauge-dependent information:*

1. The fixed-time support statement

$$A_\diamond(\mathcal{N}_{H,t}) = 0 \iff U_H(t)^T = U_H(t) \quad (125)$$

is independent of the admissible scalar normalization gauge.

2. Numerical resource values are not invariant under arbitrary normalization changes, but are quantitatively equivalent under a common strict-contraction bound according to Proposition 9.

3. The trajectory-derived resource based on the per-Hamiltonian tight normalization is invariant under positive overall energy rescaling when the full positive-time domain is optimized.

Proof. Items (1) and (2) are respectively Proposition 8 and Proposition 9. Item (3) follows from Lemma 5, introduced below. $\square$

Remark 14 (Interpretation). *The physically invariant statement supplied by the present construction is therefore a statement about transpose-reciprocity breaking and its operational detectability after a specified normalization gauge has been fixed. Claims about absolute resource magnitudes are always understood relative to that gauge.*

V. TIME-OPTIMIZED TRAJECTORY-DERIVED RESOURCE

Definition 22 (Per-Hamiltonian tight normalization gauge). *For a fixed finite-dimensional Hamiltonian H , fix $\varepsilon_0 > 0$ and define*

$$\kappa_H^{\text{tight}}(t) = (1 + \varepsilon_0) \|U_H(t)\|_\infty^2. \quad (126)$$

Then

$$K_{H,t}^{\text{tight}} = \frac{U_H(t)}{\sqrt{\kappa_H^{\text{tight}}(t)}} \quad (127)$$

satisfies

$$\|K_{H,t}^{\text{tight}}\|_\infty = \frac{1}{\sqrt{1 + \varepsilon_0}} =: q_{\text{tight}} < 1. \quad (128)$$

For the remainder of this manuscript, $\mathcal{N}_{H,t}^{\text{tight}}$ denotes the canonical Julia channel generated by this gauge.

Definition 23 (Trajectory-derived resources). *For a Hamiltonian H and an operational time domain $\mathcal{T} \subset (0, \infty)$, define the gauge-dependent trajectory resources*

$$A_{\diamond, \kappa}^{\text{sup}}(H; \mathcal{T}) = \sup_{t \in \mathcal{T}} A_{\diamond}(\mathcal{N}_{H,t}^{(\kappa)}), \quad (129)$$

and

$$R_{\text{rob}, \kappa}^{\text{sup}}(H; \mathcal{T}) = \sup_{t \in \mathcal{T}} R_{\text{rob}}(\mathcal{N}_{H,t}^{(\kappa)}). \quad (130)$$

When the tight gauge of Definition 22 is used and $\mathcal{T} = (0, \infty)$, we abbreviate

$$A_{\diamond}^{\text{sup}}(H) = A_{\diamond, \text{tight}}^{\text{sup}}(H; (0, \infty)), \quad R_{\text{rob}}^{\text{sup}}(H) = R_{\text{rob}, \text{tight}}^{\text{sup}}(H; (0, \infty)). \quad (131)$$

Lemma 5 (Exact scale covariance of the tight gauge). *For every $\lambda > 0$,*

$$\mathcal{N}_{\lambda H, t}^{\text{tight}} = \mathcal{N}_{H, \lambda t}^{\text{tight}} \quad (132)$$

Consequently,

$$A_{\diamond}^{\text{sup}}(\lambda H) = A_{\diamond}^{\text{sup}}(H), \quad R_{\text{rob}}^{\text{sup}}(\lambda H) = R_{\text{rob}}^{\text{sup}}(H). \quad (133)$$

Proof. Since

$$U_{\lambda H}(t) = U_H(\lambda t), \quad (134)$$

one has

$$\|U_{\lambda H}(t)\|_{\infty} = \|U_H(\lambda t)\|_{\infty}, \quad (135)$$

and therefore

$$\kappa_{\lambda H}^{\text{tight}}(t) = \kappa_H^{\text{tight}}(\lambda t). \quad (136)$$

Hence

$$K_{\lambda H, t}^{\text{tight}} = K_{H, \lambda t}^{\text{tight}}, \quad (137)$$

which gives the first identity. The second follows by the bijection $s = \lambda t$ of $(0, \infty)$. $\square$

Definition 24 (Pointwise trajectory-level free transformation). *Let $\mathcal{T} \subset (0, \infty)$ be an operational time domain and let $\mathfrak{N} = \{\mathcal{N}_t\}_{t \in \mathcal{T}}$ be a trajectory with $\mathcal{N}_t \in \text{Obj}_L$. A pointwise trajectory-level free transformation is a family*

$$\mathfrak{f} = \{\Theta_t\}_{t \in \mathcal{T}} \quad (138)$$

with each $\Theta_t \in \mathfrak{D}_L$ such that

$$(\mathfrak{f}\mathfrak{N})_t := \Theta_t(\mathcal{N}_t), \quad t \in \mathcal{T}, \quad (139)$$

and such that the family preserves the relevant object class and has uniform locality bounds on every compact subinterval of $\mathcal{T}$. No independent time reparametrization is

included in $\mathfrak{f}$; time reparametrization is treated separately through the exact scale covariance of Lemma 5.

This is deliberately a pointwise derived construction, not a claim of a causal process-supermap or process-tensor resource theory. A single physically time-consistent control protocol would require additional causal compatibility conditions between the maps at different times.

Theorem 4 (Trajectory-level monotonicity). *For every trajectory-level free transformation $\mathfrak{f}$,*

$$\sup_{t \in \mathcal{T}} A_{\diamond}(\Theta_t(\mathcal{N}_t)) \leq \sup_{t \in \mathcal{T}} A_{\diamond}(\mathcal{N}_t), \quad (140)$$

and

$$\sup_{t \in \mathcal{T}} R_{\text{rob}}(\Theta_t(\mathcal{N}_t)) \leq \sup_{t \in \mathcal{T}} R_{\text{rob}}(\mathcal{N}_t). \quad (141)$$

Proof. The fixed-time monotonicity theorems apply pointwise in t , and taking the supremum preserves the inequalities. $\square$

Proposition 11 (No fixed-time resource persistence law from skin length alone). *For the uniform Hatano–Nelson family, scaling $H_{\lambda, g} = \lambda H_g$ leaves*

$$\rho_{\text{GBZ}}(H_{\lambda, g}) = \frac{|g|}{a}, \quad \ell_{\text{skin}} = \frac{a}{|g|}, \quad (142)$$

unchanged, while

$$U_{\lambda, g}(t) = U_g(\lambda t). \quad (143)$$

Under a common fixed normalization $\kappa(t)$ independent of λ , with uniform thermodynamic bounds,

$$A_{\diamond, \infty}(H_{\lambda, g}; t) \rightarrow 0 \quad (\lambda \rightarrow 0) \quad (144)$$

at fixed operational time. Hence a fixed-time persistence number cannot be a universal quantitative function of skin length alone.

Proof. The GBZ radius is determined by the hopping ratio and is invariant under overall energy rescaling. At fixed t ,

$$\|U_{\lambda, g}(t) - \mathbb{1}\|_{\infty} \leq e^{\lambda \|H_g\|_{\infty} t} - 1 \rightarrow 0. \quad (145)$$

The associated normalized propagator converges to a scalar multiple of $\mathbb{1}$, whose Julia channel is reciprocal. Continuity of the Julia completion and of $A_{\diamond}$ therefore gives the stated vanishing limit. $\square$

VI. FREE SUPERCHANNELS, DILUTION, AND THE EXACT SCOPE OF CONVERSION

Proposition 12 (No-ancilla covariant realization). *Let $\Gamma \in \text{Obj}_L$ and define*

$$\Theta_{\Gamma}(\mathcal{N}) = \Gamma^{\text{rev}} \circ \mathcal{N} \circ \Gamma. \quad (146)$$

Then Θ_{Γ} is a physical free superchannel.

Proof. Bistochasticity of Γ implies bistochasticity of Γ^{rev} . Quasi-locality and object preservation follow from $\Gamma \in \text{Obj}_L$ and Lemma 3. By Lemma 4,

$$[\Theta_\Gamma(\mathcal{N})]^{\text{rev}} = (\Gamma^{\text{rev}} \circ \mathcal{N} \circ \Gamma)^{\text{rev}} \quad (147)$$

$$= \Gamma^{\text{rev}} \circ \mathcal{N}^{\text{rev}} \circ \Gamma \quad (148)$$

$$= \Theta_\Gamma(\mathcal{N}^{\text{rev}}). \quad (149)$$

Thus the covariance condition holds. $\square$

Proposition 13 (Covariant dilution by a reciprocal branch). *Fix $\mathcal{R}_0 \in \mathfrak{F}_L$ and $p \in [0, 1]$. Suppose the laboratory control model admits a deterministic randomized switch that applies $\mathcal{N}$ with probability p and the fixed reciprocal channel $\mathcal{R}_0$ with probability $1-p$, with no object-class violation. Then*

$$\Theta_p(\mathcal{N}) = p\mathcal{N} + (1-p)\mathcal{R}_0 \quad (150)$$

is a free superchannel and

$$A_\diamond(\Theta_p(\mathcal{N})) = pA_\diamond(\mathcal{N}), \quad R_{\text{rob}}(\Theta_p(\mathcal{N})) \leq pR_{\text{rob}}(\mathcal{N}). \quad (151)$$

Proof. The covariance and asymmetry identity follow by linearity. If

$$\frac{\mathcal{N} + s\mathcal{M}}{1+s} = \mathcal{F} \in \mathfrak{F}_L, \quad (152)$$

then

$$\Theta_p(\mathcal{N}) + ps\mathcal{M} = p(1+s)\mathcal{F} + (1-p)\mathcal{R}_0, \quad (153)$$

which is free after division by $1+ps$. Hence ps is feasible, and taking the infimum gives the robustness bound. $\square$

Definition 25 (Ancilla-free superchannel). *A superchannel is ancilla free if it has the form*

$$\Theta(\mathcal{N}) = \Gamma_{\text{post}} \circ \mathcal{N} \circ \Gamma_{\text{pre}} \quad (154)$$

with no memory system. The main QRT does not assume that arbitrary ancilla-free covariant superchannels can be classified from covariance restricted to Obj_L .

Definition 26 (Unitary-congruence free subclass). *For a unitary V let $\mathcal{V}(X) = VXV^\dagger$. Define*

$$\boxed{\Theta_V(\mathcal{N}) = \mathcal{V}^{\text{rev}} \circ \mathcal{N} \circ \mathcal{V}, \quad \mathcal{V}^{\text{rev}}(X) = V^T X V^*}. \quad (155)$$

For each finite L , $\mathfrak{D}_{\text{uc}}(L)$ denotes the set of such Θ_V that preserve Obj_L ; for thermodynamic families we additionally require uniform quasi-locality of the implementing unitaries. Every member is a deterministic, ancilla-free, physical free superchannel.

Remark 15. The full ancilla-assisted covariant superchannel class, and even the full ancilla-free covariant class beyond $\mathfrak{D}_{\text{uc}}(L)$, are not classified here. In particular, deterministic attenuation $\mathcal{N}_g \rightarrow \mathcal{N}_{g'}$ for $|g'| < |g|$ is not inferred from the monotones. Any sufficiency statement for the full class requires an explicit physical protocol.

VII. CONCRETE HATANO–NELSON CHANNEL CALIBRATION

Consider the two-site calibration Hamiltonian

$$H_0 = m\mathbb{1} + t_0(|1\rangle\langle 2| + |2\rangle\langle 1|), \quad (156)$$

and the gauge family

$$H_g = S_g H_0 S_g^{-1}, \quad S_g|1\rangle = |1\rangle, \quad S_g|2\rangle = e^{-g}|2\rangle, \quad (157)$$

so that

$$H_g = m\mathbb{1} + t_0(e^g|1\rangle\langle 2| + e^{-g}|2\rangle\langle 1|). \quad (158)$$

Throughout this calibration, we assume $m, t_0 \in \mathbb{R}$ with $t_0 > 0$.

At

$$\tau_* = \frac{\pi}{2t_0}, \quad (159)$$

one has, up to the global phase $e^{-im\tau_*}$,

$$U_g(\tau_*) = -i \begin{pmatrix} 0 & e^g \\ e^{-g} & 0 \end{pmatrix}, \quad \|U_g(\tau_*)\|_\infty = e^{|g|}. \quad (160)$$

For a common normalization $\kappa > e^{2g_{\text{max}}}$ on $|g| \leq g_{\text{max}}$, set

$$u = \kappa^{-1/2}. \quad (161)$$

Then

$$K_g = -iu \begin{pmatrix} 0 & e^g \\ e^{-g} & 0 \end{pmatrix}, \quad (162)$$

and the Julia defects are

$$D_K = \text{diag}(a, b), \quad D_{K^\dagger} = \text{diag}(b, a), \quad (163)$$

with

$$a = \sqrt{1 - u^2 e^{-2g}}, \quad b = \sqrt{1 - u^2 e^{2g}}. \quad (164)$$

Theorem 5 (Exact two-site channel asymmetry). *For the two-site Hatano–Nelson calibration at τ_* ,*

$$\boxed{A_\diamond(\mathcal{N}_{g, \tau_*}) = \sqrt{1 - c(g)^2}}, \quad (165)$$

where

$$\boxed{c(g) = u^2 + \sqrt{(1 - u^2 e^{2g})(1 - u^2 e^{-2g})}}. \quad (166)$$

Moreover, for $g > 0$,

$$\boxed{\frac{d}{dg} A_\diamond(\mathcal{N}_{g, \tau_*}) > 0}. \quad (167)$$

Proof. The relative unitary for discrimination of $\tilde{\mathcal{E}}_{K_g}$ and $\tilde{\mathcal{E}}_{K_g}^{\text{rev}}$ is

$$W_g = \tilde{U}_{K_g}^\dagger \tilde{U}_{K_g}^T. \quad (168)$$

For the calibration unitary, direct block multiplication gives that the spectrum of W_g consists of $e^{\pm i\theta(g)}$, each with multiplicity two, with

$$\cos \theta(g) = u^2 + ab = c(g). \quad (169)$$

For unitary channels, the exact discrimination formula gives

$$\frac{1}{2} \|\mathcal{U} - \mathcal{V}\|_\diamond = \sqrt{1 - \nu(W)^2}, \quad (170)$$

where

$$\nu(W) = \min_{\|\psi\|=1} |\langle \psi | W | \psi \rangle|. \quad (171)$$

Because the spectrum is $e^{\pm i\theta}$ with $0 \leq \theta \leq \pi/2$ in the strict calibration regime, the numerical range is the line segment between these two eigenvalues and $\nu(W) = \cos \theta = c(g)$. Hence

$$A_\diamond = \sqrt{1 - c(g)^2}. \quad (172)$$

Writing $y = e^{2g} > 1$ gives

$$c(g) = u^2 + \sqrt{1 + u^4 - u^2(y + y^{-1})}, \quad (173)$$

hence

$$c'(g) = -\frac{u^2(y - y^{-1})}{\sqrt{1 + u^4 - u^2(y + y^{-1})}} < 0. \quad (174)$$

Therefore $A'_\diamond(g) > 0$. $\square$

Corollary 3 (Exact scalar ordering on the calibration orbit). *For the common normalization of Theorem 5,*

$$|g| \geq |g'| \iff A_\diamond(\mathcal{N}_{g,\tau_*}) \geq A_\diamond(\mathcal{N}_{g',\tau_*}). \quad (175)$$

Consequently every deterministic free conversion $\mathcal{N}_{g,\tau_*} \succeq_\Omega \mathcal{N}_{g',\tau_*}$ must satisfy

$$|g| \geq |g'|. \quad (176)$$

Proof. The first statement follows from evenness in g and strict monotonicity in $|g|$. The second follows from monotonicity of $A_\diamond$ under free superchannels. $\square$

Lemma 6 (Phase-invariant unitary-congruence diagnostic). *For a unitary U acting on a d -dimensional space define the unitary-congruence cosquare*

$$C_{\text{cs}}(U) = U^*U, \quad \chi_{\text{cs}}(U) = \frac{1}{d^2} |\text{Tr}(U^*U)|^2. \quad (177)$$

Under

$$U \mapsto e^{i\phi} V^T U V, \quad (178)$$

one has

$$\chi_{\text{cs}}(e^{i\phi} V^T U V) = \chi_{\text{cs}}(U). \quad (179)$$

For the two-site Julia unitary at τ_ ,*

$$\chi_{\text{cs}}(\tilde{U}_{K_g}) = c(g)^2, \quad (180)$$

with $c(g)$ from Theorem 5. Since $c(g) > 0$ and is strictly decreasing in $|g|$ for $|g| > 0$, χ_{cs} is strictly decreasing in $|g|$ on the calibration orbit.

Proof. Let

$$U' = e^{i\phi} V^T U V. \quad (181)$$

Then

$$(U')^* U' = V^\dagger U^* (V^* V^T) U V = V^\dagger (U^* U) V, \quad (182)$$

so C_{cs} is unitarily similar to the original cosquare and its trace is invariant; the phase cancels exactly. For a unitary U ,

$$U^* U = (U^\dagger U^T)^\dagger, \quad (183)$$

so it has the complex-conjugate eigenvalue multiset of the discrimination relative unitary. In the present calibration the spectrum is the conjugate-symmetric set $e^{\pm i\theta}$, each twice, and therefore

$$\text{Tr}(\tilde{U}_{K_g}^* \tilde{U}_{K_g}) = 4 \cos \theta(g) = 4c(g). \quad (184)$$

The four-dimensional normalization gives $\chi_{\text{cs}} = c(g)^2$. $\square$

Theorem 6 (No nontrivial conversion in the ancilla-free unitary-congruence subclass). *Suppose $\Theta_V \in \mathfrak{D}_{\text{uc}}(L)$ maps $\mathcal{N}_{g,\tau_*}$ to $\mathcal{N}_{g',\tau_*}$. Then*

$$|g| = |g'|. \quad (185)$$

Thus no nontrivial attenuation or amplification occurs inside this explicitly defined ancilla-free unitary-congruence subclass.

Proof. Equality of the output unitary channels implies that their implementing unitaries differ only by a global phase, so

$$\tilde{U}_{K_{g'}} = e^{i\phi} V^T \tilde{U}_{K_g} V \quad (186)$$

for some phase ϕ . Lemma 6 therefore gives

$$c(g')^2 = c(g)^2. \quad (187)$$

Both $c(g)$ and $c(g')$ are positive, so $c(g') = c(g)$; strict monotonicity in $|g|$ then implies $|g| = |g'|$. $\square$

Remark 16 (Status of the full conversion problem). *The rigorous conversion information established here has three distinct layers. First, the monotone $A_\diamond$ gives the necessary condition $|g| \geq |g'|$ for any free conversion on the calibration orbit. Second, the explicitly defined subclass $\mathfrak{D}_{\text{uc}}(L)$ is rigid: within it, $|g|$ cannot change at all. Third, the full ancilla-assisted preorder remains open. In particular, the sufficient direction*

$$|g| > |g'| \implies \mathcal{N}_{g, \tau_*} \succeq_{\mathfrak{D}} \mathcal{N}_{g', \tau_*} \quad (188)$$

is not claimed and cannot be inferred from monotonicity alone. Establishing it requires an explicit deterministic attenuation protocol in the full free-operation model. Sec. XVII isolates a concrete obstruction facing the most natural candidate protocol and states this sufficiency direction as a precise, numerically testable conjecture.

VIII. EXACT SCOPE OF THE NHSE-RESOURCE CORRESPONDENCE

We stress at the outset of this section that the channel resource constructed above is, and remains, a detector of transpose-reciprocity breaking as an algebraic property of H . It is *not* a general-purpose detector of the physical phenomenon of non-Hermitian skin accumulation. Conventional non-Hermitian skin accumulation can arise through mechanisms that do not require $H^T \neq H$ (e.g. genuinely non-Abelian or Lindbladian mechanisms discussed in Sec. XXII), so no universal equivalence between $A_\diamond^{\text{sup}} > 0$ and “NHSE occurs” is asserted or should be inferred from what follows. The exact logical scope of the correspondence is stated in three layers: a universal statement about transpose-reciprocity breaking (unconditional), a one-way witness statement for the sector where conventional NHSE happens to coincide with transpose-reciprocity breaking, and a biconditional statement that holds only on model classes explicitly shown to satisfy Definition 30. An explicit exact example established here is the uniform Hatano–Nelson family (Proposition 26), where the equivalence is close to definitional: the GBZ radius $\rho_{\text{GBZ}} = |g|/a$ on that orbit is parametrized by the same g that controls $H_g^T \neq H_g$. The resulting biconditional should therefore be read as an exact solved calibration, not as evidence for a broader universal correspondence.

Definition 27 (Transpose-reciprocity-breaking class). *Define*

$$\mathcal{C}_{\text{TRB}} = \{H : H^T \neq H\}. \quad (189)$$

Definition 28 (Conventional NHSE class). *Let $\mathcal{C}_{\text{NHSE}}^{\text{conv}}$ denote a class of one-dimensional finite-range Hamiltonians with open boundaries satisfying the conventional extensive skin criterion: there exists an extensive bulk spectral window I and a sequence $\ell_L = o(L)$ such that*

$$\limsup_{L \rightarrow \infty} C_I^{\text{edge}}(L, \ell_L) > 0. \quad (190)$$

The definition is based on OBC eigenstate concentration and does not by itself impose transpose reciprocity breaking.

Definition 29 (Transpose-reciprocity-breaking conventional NHSE class). *Define*

$$\mathcal{C}_{\text{TR-NHSE}} = \mathcal{C}_{\text{TRB}} \cap \mathcal{C}_{\text{NHSE}}^{\text{conv}}. \quad (191)$$

Thus membership in $\mathcal{C}_{\text{TR-NHSE}}$ explicitly includes both conventional OBC skin accumulation and transpose-reciprocity breaking.

Theorem 7 (Exact trajectory-level characterization of transpose-reciprocity breaking). *Let H be finite dimensional and let the operational time domain $\mathcal{T}$ contain a sequence*

$$t_j \downarrow 0. \quad (192)$$

For any admissible strict scalar normalization κ , define

$$A_{\diamond, \kappa}^{\text{sup}}(H; \mathcal{T}) = \sup_{t \in \mathcal{T}} A_\diamond(\mathcal{N}_{H, t}^{(\kappa)}). \quad (193)$$

Then

$$A_{\diamond, \kappa}^{\text{sup}}(H; \mathcal{T}) > 0 \iff H^T \neq H. \quad (194)$$

Equivalently,

$$A_{\diamond, \kappa}^{\text{sup}}(H; \mathcal{T}) = 0 \iff H^T = H. \quad (195)$$

Proof. If $H^T = H$, then

$$U_H(t)^T = U_H(t) \quad (196)$$

for every t , and Proposition 2 gives

$$A_\diamond(\mathcal{N}_{H, t}^{(\kappa)}) = 0 \quad (197)$$

for every t . Hence the supremum vanishes.

Conversely, suppose $H^T \neq H$. The short-time expansion gives

$$U_H(t)^T - U_H(t) = -i(H^T - H)t + O(t^2). \quad (198)$$

Since $H^T - H \neq 0$, the analytic matrix-valued function $t \mapsto U_H(t)^T - U_H(t)$ cannot vanish identically in any neighborhood of $t = 0$. Therefore there exists a sequence $t_j \downarrow 0$ for which

$$U_H(t_j)^T \neq U_H(t_j). \quad (199)$$

By Proposition 8,

$$A_\diamond(\mathcal{N}_{H, t_j}^{(\kappa)}) > 0, \quad (200)$$

and hence

$$A_{\diamond, \kappa}^{\text{sup}}(H; \mathcal{T}) > 0. \quad (201)$$

□

Corollary 4 (One-way resource witness for the transpose-breaking NHSE sector). *For every*

$$H \in \mathcal{C}_{\text{TR-NHSE}}, \quad (202)$$

one has

$$\boxed{A_{\diamond}^{\text{sup}}(H) > 0.} \quad (203)$$

Thus positivity of the trajectory-derived channel resource is a necessary operational consequence of membership in the transpose-reciprocity-breaking sector of conventional NHSE.

Proof. By definition, $H \in \mathcal{C}_{\text{TR-NHSE}}$ implies $H^T \neq H$. Theorem 7 then gives $A_{\diamond}^{\text{sup}}(H) > 0$ for the tight gauge. $\square$

Remark 17 (What is and is not claimed by the witness theorem). *Theorem 7 is a theorem about transpose-reciprocity breaking, not a theorem that every Hamiltonian exhibiting conventional NHSE must satisfy $H^T \neq H$. Consequently, the implication*

$$\text{conventional NHSE} \implies A_{\diamond}^{\text{sup}} > 0 \quad (204)$$

is asserted here only for the explicitly defined sector $\mathcal{C}_{\text{TR-NHSE}}$. No universal converse for the full set $\mathcal{C}_{\text{NHSE}}^{\text{conv}}$ is claimed.

Definition 30 (Transpose-reciprocity-complete NHSE class). *A model class $\mathcal{C}$ is called transpose-reciprocity complete with respect to conventional NHSE if*

$$\boxed{H \in \mathcal{C} \implies [H \in \mathcal{C}_{\text{NHSE}}^{\text{conv}} \iff H^T \neq H].} \quad (205)$$

Equivalently, within $\mathcal{C}$, conventional OBC skin accumulation and transpose-reciprocity breaking define the same subset.

Proposition 14 (Transfer of exact NHSE characterization to complete model classes). *Let $\mathcal{C}$ be a transpose-reciprocity-complete NHSE class in the sense of Definition 30. Suppose its operational time domain contains a sequence $t_j \downarrow 0$, and let the channel resource be defined with any admissible strict scalar normalization or with the per-Hamiltonian tight normalization. Then*

$$\boxed{H \in \mathcal{C} \implies [A_{\diamond}^{\text{sup}}(H) > 0 \iff H \in \mathcal{C}_{\text{NHSE}}^{\text{conv}}].} \quad (206)$$

Thus the channel resource is an exact detector of conventional NHSE within the stated complete model class.

Proof. By Theorem 7,

$$A_{\diamond}^{\text{sup}}(H) > 0 \iff H^T \neq H. \quad (207)$$

By the defining property of the class $\mathcal{C}$,

$$H^T \neq H \iff H \in \mathcal{C}_{\text{NHSE}}^{\text{conv}}. \quad (208)$$

Combining the two equivalences proves the result. $\square$

Remark 18 (Logical hierarchy). *The preceding results have three distinct logical strengths:*

$$H^T \neq H \implies A_{\diamond, \kappa}^{\text{sup}}(H; \mathcal{T}) > 0, \quad (209)$$

$$H \in \mathcal{C}_{\text{TR-NHSE}} \implies A_{\diamond}^{\text{sup}}(H) > 0, \quad (210)$$

$$H \in \mathcal{C}, \quad \mathcal{C} \text{ complete} \implies [A_{\diamond}^{\text{sup}}(H) > 0 \iff H \in \mathcal{C}_{\text{NHSE}}^{\text{conv}}]. \quad (211)$$

Equation (209) is universal within the finite-dimensional transpose-breaking channel architecture. Equation (210) is the one-way NHSE witness statement. Equation (211) is the class-conditional exact correspondence. No stronger universal NHSE equivalence is used elsewhere in the paper.

IX. DIAGNOSTIC SECTOR I: NON-BLOCH SPECTRAL STRUCTURE

The channel QRT is the primitive theory. Spectral quantities below are structured Hamiltonian diagnostics, not additional free-operation monotones.

For a translation-invariant one-dimensional lattice,

$$H(\beta) = \sum_{r=-p}^q H_r \beta^r. \quad (212)$$

For a fixed energy E , let the roots of

$$\det[H(\beta) - E] = 0 \quad (213)$$

be ordered as

$$|\beta_1| \leq \dots \leq |\beta_{2M}|. \quad (214)$$

For nearest-neighbor M -channel models, the GBZ condition is

$$|\beta_M| = |\beta_{M+1}|. \quad (215)$$

Definition 31 (GBZ radius witness). *Define*

$$\boxed{\rho_{\text{GBZ}}(H) = \frac{1}{a} \sup_{\beta \in \mathcal{C}_{\text{GBZ}}} |\ln |\beta||.} \quad (216)$$

Lemma 7 (Reciprocal root pairing). *Consider a clean reciprocal nearest-neighbor chain with invertible intercell hopping blocks and*

$$H_{-1} = H_1^T, \quad H_0 = H_0^T. \quad (217)$$

Then

$$H(\beta)^T = H(\beta^{-1}), \quad (218)$$

and

$$\boxed{\det[H(\beta) - E] = \det[H(\beta^{-1}) - E].} \quad (219)$$

Thus nonzero roots occur in reciprocal pairs

$$\boxed{\beta \leftrightarrow \beta^{-1}.} \quad (220)$$

Proof. Direct substitution gives

$$H(\beta)^T = H_{-1}^T \beta^{-1} + H_0^T + H_1^T \beta = H_1 \beta^{-1} + H_0 + H_{-1} \beta = H(\beta^{-1}). \quad (221)$$

Taking determinants proves the result. $\square$

Theorem 8 (Reciprocity forces zero determinant winding). *For a clean reciprocal chain and a reference energy E in a point gap,*

$$W_{\text{BZ}}(E) = \frac{1}{2\pi i} \int_{-\pi}^{\pi} dk \partial_k \log \det[H(k) - E] = 0. \quad (222)$$

Proof. Reciprocity implies $H(-k) = H(k)^T$, hence

$$\det[H(-k) - E] = \det[H(k) - E]. \quad (223)$$

Thus the logarithmic derivative is odd on the Brillouin zone and the integral vanishes. $\square$

Corollary 5 (Reciprocal GBZ is the unit circle). *Under the hypotheses of Lemma 7,*

$$|\beta_M| = |\beta_{M+1}| = 1, \quad (224)$$

and therefore

$$\rho_{\text{GBZ}} = 0. \quad (225)$$

Proof. Reciprocal pairing gives

$$|\beta_j| |\beta_{2M+1-j}| = 1. \quad (226)$$

At the GBZ crossing, the two middle moduli are equal, so both are unity. $\square$

Proposition 15 (Hatano–Nelson spectral calibration). *For*

$$H_g(\beta) = m + J e^g \beta + J e^{-g} \beta^{-1}, \quad g \in \mathbb{R}, \quad (227)$$

the two roots of $H_g(\beta) = E$ satisfy

$$\beta_1 \beta_2 = e^{-2g}, \quad (228)$$

and the scalar GBZ radius is

$$\rho_{\text{GBZ}} = \frac{|g|}{a}. \quad (229)$$

Proof. Multiplying the characteristic equation by β gives a quadratic whose leading and constant coefficients differ by $e^{\pm g}$, so the product of its roots is e^{-2g} . The scalar GBZ condition forces equal root moduli and therefore $|\beta_{\text{GBZ}}| = e^{-|g|}$. $\square$

X. NECESSITY AND UNIFORM SHORT-TIME BRIDGE

Assumption 1 (Clean finite-range class). *H is a one-dimensional translation-invariant finite-range single-particle Hamiltonian with internal dimension M . For GBZ root-pairing conclusions we additionally assume nearest-neighbor intercell blocks and their invertibility.*

Definition 32 (Real-space reciprocity). *For a translation-invariant Hamiltonian,*

$$H_r = H_{-r}^T \quad \text{for all } r. \quad (230)$$

Equivalently, the finite-size matrix satisfies $H^T = H$ in the fixed spatial basis.

Corollary 6 (Reciprocity is a necessary obstruction to conventional reciprocal-breaking NHSE). *Within the clean class,*

$$H^T = H \implies A_{\diamond}(\mathcal{N}_{H,t}) = 0, \quad W_{\text{BZ}}(E) = 0, \quad (231)$$

and under the nearest-neighbor assumptions,

$$\rho_{\text{GBZ}} = 0. \quad (232)$$

Therefore every conventional NHSE mechanism in the present transpose-reciprocity-breaking sector must satisfy

$$\text{conventional transpose-reciprocity-breaking NHSE} \implies \exists t : A_{\diamond}(\mathcal{N}_H) = 0. \quad (233)$$

Proof. The channel statement follows from Proposition 2; the spectral statements follow from Sec. IX. The final implication follows from Theorem 7 on the stated sector. $\square$

Definition 33 (Transition-probability asymmetry defect).

$$D_{\text{amp}}(H) = \max_m \sum_{n \neq m} ||H_{nm}|^2 - |H_{mn}|^2|. \quad (234)$$

A. Uniform thermodynamic short-time control

The thermodynamic short-time statement requires more than a pointwise expansion at each finite system size. To prevent the remainder from growing with L , the small-time estimate is formulated with an explicit system-size-independent bound.

Assumption 2 (Uniformly bounded thermodynamic Hamiltonian family). *Let $\{H_L\}_{L \geq 1}$ be a family of finite-dimensional Hamiltonians satisfying*

$$\sup_{L \geq 1} \|H_L\|_{\infty} \leq h < \infty. \quad (235)$$

For the thermodynamic NHSE applications below, we additionally assume uniformly bounded finite interaction range.

Lemma 8 (Uniform exponential remainder for the propagator). *Under Assumption 2,*

$$U_L(t) = \mathbb{1} - iH_L t + R_L(t), \quad (236)$$

with

$$\|R_L(t)\|_\infty \leq \frac{1}{2} h^2 t^2 e^{ht} \quad (237)$$

for every L and $t \geq 0$.

Proof. Using the exponential series,

$$R_L(t) = \sum_{r=2}^{\infty} \frac{(-itH_L)^r}{r!}, \quad (238)$$

$$\|R_L(t)\|_\infty \leq \sum_{r=2}^{\infty} \frac{(ht)^r}{r!} \leq \frac{1}{2} h^2 t^2 e^{ht}. \quad (239)$$

The bound is independent of L . $\square$

Lemma 9 (Uniform short-time control of transition intensities). *Under Assumption 2, for every m and every $t \geq 0$,*

$$\sum_{n \neq m} |U_L(t)_{nm}|^2 - t^2 |(H_L)_{nm}|^2 \leq \mathcal{E}_h(t), \quad (240)$$

and

$$\sum_{n \neq m} |U_L(t)_{mn}|^2 - t^2 |(H_L)_{mn}|^2 \leq \mathcal{E}_h(t), \quad (241)$$

where

$$\mathcal{E}_h(t) = h^3 t^3 e^{ht} + \frac{1}{4} h^4 t^4 e^{2ht}. \quad (242)$$

Consequently,

$$\max_m \sum_{n \neq m} |U_L(t)_{nm}|^2 - |U_L(t)_{mn}|^2 - t^2 (|(H_L)_{nm}|^2 - |(H_L)_{mn}|^2) \leq 2\mathcal{E}_h(t). \quad (243)$$

All constants are independent of L .

Proof. Write

$$U_L(t)_{nm} = -it(H_L)_{nm} + R_L(t)_{nm}. \quad (244)$$

Then

$$|U_L(t)_{nm}|^2 - t^2 |(H_L)_{nm}|^2 \leq 2t |(H_L)_{nm}| |R_L(t)_{nm}| + |R_L(t)_{nm}|^2. \quad (245)$$

Summing and using Cauchy–Schwarz gives

$$\sum_n 2t |(H_L)_{nm}| |R_L(t)_{nm}| \leq 2t \left(\sum_n |(H_L)_{nm}|^2 \right)^{1/2} \left(\sum_n |R_L(t)_{nm}|^2 \right)^{1/2}, \quad (246)$$

$$\leq 2th \|R_L(t)\|_\infty \quad (247)$$

$$\leq h^3 t^3 e^{ht}, \quad (248)$$

while

$$\sum_n |R_L(t)_{nm}|^2 \leq \|R_L(t)\|_\infty^2 \leq \frac{1}{4} h^4 t^4 e^{2ht}. \quad (249)$$

This proves the first estimate. The row estimate follows by applying the same argument to $U_L(t)^\dagger$ and $H_L^\dagger$. Combining the two estimates gives the final inequality. $\square$

Proposition 16 (Uniform finite-size short-time resource lower bound). *Let $\{H_L\}$ satisfy Assumption 2, and let $\kappa_L(t)$ be an admissible scalar normalization gauge satisfying*

$$\kappa_L(t) = \kappa_0 + O(t) \quad (250)$$

uniformly in L , with $\kappa_0 > 0$. Then there exist $t_0 > 0$ and $C_ > 0$, independent of L , such that for all $0 < t < t_0$,*

$$A_\diamond(\mathcal{N}_{H_L,t}) \geq \frac{t^2}{2\kappa_0} D_{\text{amp}}(H_L) - C_* t^3. \quad (251)$$

Thus the leading short-time lower-bound coefficient is uniform in system size.

Proof. By Proposition 24,

$$A_\diamond(\mathcal{N}_{H_L,t}) \geq \frac{1}{2\kappa_L(t)} \max_m \sum_n |U_L(t)_{nm}|^2 - |U_L(t)_{mn}|^2. \quad (252)$$

The diagonal term $n = m$ vanishes identically. Hence Lemma 9 gives

$$\max_m \sum_{n \neq m} |U_L(t)_{nm}|^2 - |U_L(t)_{mn}|^2 \geq t^2 D_{\text{amp}}(H_L) - 2\mathcal{E}_h(t). \quad (253)$$

Because

$$\kappa_L(t)^{-1} = \kappa_0^{-1} + O(t) \quad (254)$$

uniformly in L , and $D_{\text{amp}}(H_L)$ is uniformly bounded for a finite-range uniformly bounded family, the normalization correction contributes only an L -independent $O(t^3)$ term. The remainder $2\mathcal{E}_h(t)$ is likewise $O(t^3)$ uniformly in L . Collecting constants proves the stated lower bound. $\square$

Corollary 7 (Thermodynamic small-time lower bound).

Under the hypotheses of Proposition 16,

$$\liminf_{t \rightarrow 0^+} \frac{A_{\diamond,\infty}(t)}{t^2} \geq \frac{1}{2\kappa_0} \limsup_{L \rightarrow \infty} D_{\text{amp}}(H_L). \quad (255)$$

$$\limsup_{L \rightarrow \infty} D_{\text{amp}}(H_L) > 0, \quad (256)$$

then the thermodynamic channel resource is strictly positive at sufficiently small nonzero times.

Proof. From Proposition 16,

$$A_{\diamond}(\mathcal{N}_{H_L,t}) \geq \frac{t^2}{2\kappa_0} D_{\text{amp}}(H_L) - C_* t^3. \quad (257)$$

Taking $\limsup_{L \rightarrow \infty}$ gives

$$A_{\diamond,\infty}(t) \geq \frac{t^2}{2\kappa_0} \limsup_{L \rightarrow \infty} D_{\text{amp}}(H_L) - C_* t^3. \quad (258)$$

Divide by t^2 and take $\liminf_{t \rightarrow 0^+}$. $\square$

Proposition 17 (Uniform thermodynamic persistence of the Hatano–Nelson channel resource). *Consider the open uniform Hatano–Nelson family*

$$H_{g,L} = m\mathbb{1} + Je^g T_+ + Je^{-g} T_-, \quad T_- = T_+^T, \quad (259)$$

with fixed $J > 0$, $g \in \mathbb{R}$, and system-independent m, J, g . Then

$$\|H_{g,L}\|_{\infty} \leq |m| + 2J \cosh |g| =: h_g \quad (260)$$

for every L . Moreover, for all $L \geq 3$,

$$D_{\text{amp}}(H_{g,L}) = 4J^2 \sinh(2|g|). \quad (261)$$

Let the normalization satisfy

$$\kappa_L(t) = \kappa_0 + O(t) \quad (262)$$

uniformly in L , with $\kappa_0 > 0$. Then

$$A_{\diamond,\infty}(t) \geq \frac{2J^2 \sinh(2|g|)}{\kappa_0} t^2 - C_g t^3 \quad (263)$$

for sufficiently small t , where C_g is finite and independent of L . Consequently,

$$\liminf_{t \rightarrow 0^+} \frac{A_{\diamond,\infty}(t)}{t^2} \geq \frac{2J^2 \sinh(2|g|)}{\kappa_0}. \quad (264)$$

Hence, for every $g \neq 0$,

$$A_{\diamond,\infty}(t) > 0 \quad (265)$$

for all sufficiently small $t > 0$.

Proof. Because $\|T_{\pm}\|_{\infty} = 1$,

$$\|H_{g,L}\|_{\infty} \leq |m| + Je^g + Je^{-g} \quad (266)$$

$$= |m| + 2J \cosh |g|, \quad (267)$$

uniformly in L . For a bulk site m , the two neighboring bonds contribute

$$|H_{m+1,m}|^2 - |H_{m,m+1}|^2 = J^2 |e^{2g} - e^{-2g}|, \quad (268)$$

$$|H_{m-1,m}|^2 - |H_{m,m-1}|^2 = J^2 |e^{2g} - e^{-2g}|. \quad (269)$$

Thus, for $L \geq 3$,

$$D_{\text{amp}}(H_{g,L}) = 2J^2 |e^{2g} - e^{-2g}| \quad (270)$$

$$= 4J^2 \sinh(2|g|). \quad (271)$$

The result follows from Corollary 7. $\square$

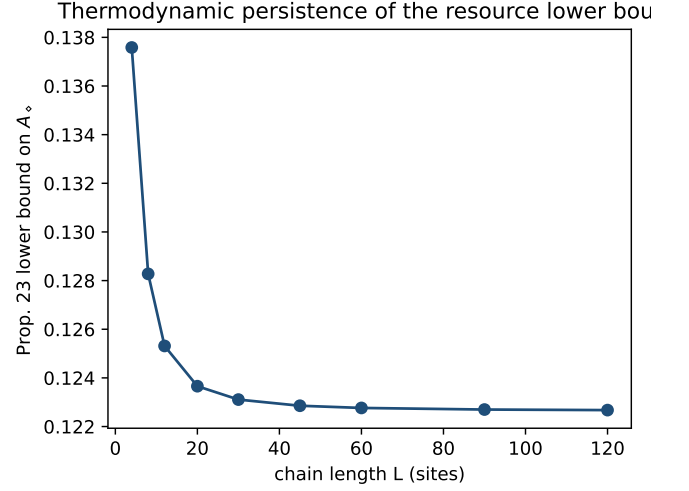


FIG. 2. Numerical evaluation of the Proposition 16 lower bound on $A_{\diamond}$, as a function of system size L for the uniform Hatano–Nelson family at fixed $g \neq 0$, using the tight normalization of Remark 20. The bound decreases monotonically from 0.1376 at $L = 4$ and converges to ≈ 0.1227 by $L = 120$, with successive differences shrinking rapidly (e.g. $L = 90 \rightarrow 120$ changes the bound by $< 3 \times 10^{-5}$). This is a clean numerical confirmation that the thermodynamic short-time lower bound of Corollary 7 converges to a finite, strictly positive value rather than degrading with system size.

Remark 19 (What is actually uniform in the thermodynamic limit). *The preceding proposition is stronger than a pointwise finite-size expansion. The coefficient of t^2 is independent of L , and the remainder is controlled by a constant depending only on the uniform Hamiltonian bound and the normalization gauge. Thus the statement $A_{\diamond,\infty}(t) > 0$ at sufficiently small nonzero t is not obtained by assuming that a finite-size error remains bounded in the thermodynamic limit; the L -uniform bound is established explicitly.*

Remark 20 (Tight normalization in the Hatano–Nelson family). *For the per-Hamiltonian tight normalization,*

$$\kappa_{H_{g,L}}^{\text{tight}}(t) = (1 + \varepsilon_0) \|U_{H_{g,L}}(t)\|_{\infty}^2, \quad (272)$$

one has

$$\lim_{t \rightarrow 0^+} \kappa_{H_{g,L}}^{\text{tight}}(t) = 1 + \varepsilon_0 \quad (273)$$

uniformly in L . Hence

$$\kappa_0 = 1 + \varepsilon_0 \quad (274)$$

and

$$\liminf_{t \rightarrow 0^+} \frac{A_{\diamond,\infty}(t)}{t^2} \geq \frac{2J^2 \sinh(2|g|)}{1 + \varepsilon_0}. \quad (275)$$

XI. MULTICHANNEL SHARPNESS: WINDING CANCELLATION

Proposition 18 (Winding-cancellation skin counterexample). *Let*

$$H_{\oplus,g}(k) = H_g(k) \oplus H_{-g}(k), \quad g \neq 0. \quad (276)$$

Then the two decoupled sectors have opposite conventional skin accumulation, while

$$\boxed{W_{\oplus,g}(E) = 0} \quad (277)$$

for every reference energy in a common point gap. At nontrivial times away from exact reciprocal points,

$$A_{\diamond}^{\sup}(H_{\oplus,g}) > 0. \quad (278)$$

Proof. Each block is a Hatano–Nelson chain, with right eigenvectors related to reciprocal standing waves by opposite diagonal similarity transforms. Thus the two sectors accumulate at opposite boundaries. Since $H_{-g}(k) = H_g(-k)$, their determinant winding numbers cancel:

$$W_{\oplus,g}(E) = W_g(E) + W_{-g}(E) = 0. \quad (279)$$

For $g \neq 0$, the direct-sum Hamiltonian is not transpose reciprocal in the fixed block basis, so Theorem 7 gives positive trajectory-derived channel resource. $\square$

Remark 21. *This construction shows why the channel resource should not be identified with a single scalar determinant winding integer.*

XII. DIAGNOSTIC SECTOR II: DISORDERED LYAPUNOV STRUCTURE

Definition 34 (Transfer cocycle and Lyapunov exponents). *Let*

$$\Psi_{n+1} = T_n(E)\Psi_n, \quad T^{(N)} = T_N \cdots T_1. \quad (280)$$

If the limits exist and

$$s_1(T^{(N)}) \geq \cdots \geq s_{2M}(T^{(N)}), \quad (281)$$

define

$$\boxed{\gamma_j(E) = \lim_{N \rightarrow \infty} \frac{1}{Na} \ln s_j(T^{(N)})}. \quad (282)$$

Proposition 19 (Boundary invariance of Lyapunov spectrum). *If*

$$T^{(N)} = A_{\partial} T^{(N)} B_{\partial} \quad (283)$$

with uniformly bounded invertible endpoint matrices, then

$$\boxed{\gamma'_j(E) = \gamma_j(E)}. \quad (284)$$

Proof. Uniformly bounded endpoint factors modify each singular value only by multiplicative constants independent of N . Dividing the logarithm by Na removes these constants. $\square$

Lemma 10 (Symplectic singular-value pairing). *Let*

$$J = \begin{pmatrix} 0 & \mathbb{1}_M \\ -\mathbb{1}_M & 0 \end{pmatrix}, \quad (285)$$

and suppose S obeys

$$S^T J S = J. \quad (286)$$

Then

$$\boxed{\sigma_j(S) \sigma_{2M+1-j}(S) = 1}. \quad (287)$$

Proof. From $S^T J S = J$,

$$S^{-1} = -J S^T J. \quad (288)$$

Since J is unitary and transpose preserves singular values, S^{-1} has the same singular values as S . The singular values of S^{-1} are $1/\sigma_{2M}, \dots, 1/\sigma_1$. $\square$

Theorem 9 (Reciprocity-induced Lyapunov pairing). *Let*

$$H_{n,n+1} = t_n^T, \quad H_{n+1,n} = t_n, \quad H_{n,n} = m_n, \quad (289)$$

with

$$m_n^T = m_n, \quad \sup_n \|t_n\| < \infty, \quad \sup_n \|t_n^{-1}\| < \infty. \quad (290)$$

Assume, in addition, that the Lyapunov limits in Definition 34 exist. Then

$$\boxed{\gamma_j(E) = -\gamma_{2M+1-j}(E)}. \quad (291)$$

Proof. Write $\Psi_n = (\chi_n, \chi_{n-1})$. The eigen-equation gives

$$t_n^T \chi_{n+1} = (E - m_n) \chi_n - t_{n-1} \chi_{n-1}, \quad (292)$$

and hence

$$T_n = \begin{pmatrix} (t_n^T)^{-1}(E - m_n) & -(t_n^T)^{-1}t_{n-1} \\ \mathbb{1} & 0 \end{pmatrix}. \quad (293)$$

The bilinear Wronskian

$$\mathcal{W}_n(\psi, \phi) = \psi_n^T t_n^T \phi_{n+1} - \phi_n^T t_n^T \psi_{n+1} \quad (294)$$

is independent of n . Therefore

$$T_n^T \Omega_{n+1} T_n = \Omega_n, \quad \Omega_n = \begin{pmatrix} 0 & -t_{n-1} \\ t_{n-1}^T & 0 \end{pmatrix}. \quad (295)$$

With

$$P_n = \text{diag}(\mathbb{1}, -t_{n-1}), \quad (296)$$

we have

$$P_n^T J P_n = \Omega_n, \quad (297)$$

so

$$\tilde{T}_n = P_{n+1} T_n P_n^{-1} \quad (298)$$

is symplectic. Lemma 10 gives singular-value pairing for products, while the uniformly bounded endpoint factors do not alter the Lyapunov limits. $\square$

Definition 35 (Imaginary-gauge spatial excess). *Suppose a reciprocal disordered backbone has Lyapunov spectrum*

$$\left\{ \pm \frac{L_j(E)}{a} \right\}_{j=1}^M, \quad 0 \leq L_1(E) \leq \dots \leq L_M(E), \quad (299)$$

and the nonreciprocal member is generated by

$$H_{g,\omega} = S_g H_{0,\omega} S_g^{-1}, \quad S_g |n\rangle = e^{-gn} |n\rangle. \quad (300)$$

Define

$$R_{\text{Lyap}}^{\text{skin}}(H_g; \mathcal{E}) = \frac{1}{a} \sup_{E \in \mathcal{E}} [|g| - L_1(E)]_+. \quad (301)$$

Proposition 20 (Clean Lyapunov–GBZ correspondence). *For a clean constant transfer matrix with nondegenerate diagonalizable spectrum,*

$$T = V \text{diag}(\beta_1^\downarrow, \dots, \beta_{2M}^\downarrow) V^{-1}, \quad |\beta_1^\downarrow| \geq \dots \geq |\beta_{2M}^\downarrow|, \quad (302)$$

one has

$$\gamma_j(E) = \frac{1}{a} \ln |\beta_j^\downarrow(E)|. \quad (303)$$

Thus

$$|\beta_M| = |\beta_{M+1}| \iff \gamma_M = \gamma_{M+1}. \quad (304)$$

Proof. The diagonalizable eigenvector factors contribute only $O(1)$ to the logarithms of singular values of T^N , while the diagonal powers contribute $N \ln |\beta_j|$. $\square$

XIII. DIAGNOSTIC SECTOR III: BOUNDARY CONCENTRATION

Let normalized right eigenvectors be $|\psi_{a,L}^R\rangle$ and define

$$p_{a,L}(n) = \frac{\sum_\alpha |\langle n, \alpha | \psi_{a,L}^R \rangle|^2}{\sum_{m,\alpha} |\langle m, \alpha | \psi_{a,L}^R \rangle|^2}. \quad (305)$$

Definition 36 (Edge concentration). *For an extensive spectral window I and boundary thickness ℓ ,*

$$C_I^L(L, \ell) = \frac{1}{N_I(L)} \sum_{a \in I} \sum_{n=1}^{\ell} p_{a,L}(n), \quad (306)$$

$$C_I^R(L, \ell) = \frac{1}{N_I(L)} \sum_{a \in I} \sum_{n=L-\ell+1}^L p_{a,L}(n), \quad (307)$$

and

$$C_I^{\text{edge}} = \max\{C_I^L, C_I^R\}, \quad \Delta_I^{\text{edge}} = |C_I^L - C_I^R|. \quad (308)$$

Conventional extensive skin accumulation means that there exists $\ell_L = o(L)$ such that

$$\limsup_{L \rightarrow \infty} C_I^{\text{edge}}(L, \ell_L) > 0. \quad (309)$$

Definition 37 (Collective decay profile). *For edge $\epsilon \in \{L, R\}$, let $P_I^\epsilon(r)$ denote the thermodynamic averaged eigenstate intensity at distance r from that edge, whenever the limit exists, and set*

$$G_I^\epsilon(r) = -\ln \frac{P_I^\epsilon(r)}{P_I^\epsilon(0)}. \quad (310)$$

Definition 38 (Scale-resolved spatial witness). *For a reference function $f(r) \rightarrow \infty$,*

$$R_f^{(I,\epsilon)}(H) = \limsup_{r \rightarrow \infty} \frac{G_I^\epsilon(r)}{f(r)}. \quad (311)$$

Theorem 10 (Lyapunov blindness to subexponential profiles). *If*

$$G(r) \rightarrow \infty, \quad \frac{G(r)}{r} \rightarrow 0, \quad (312)$$

then

$$\limsup_{r \rightarrow \infty} \frac{G(r)}{r} = 0. \quad (313)$$

Thus a vanishing linear decay coefficient does not imply absence of edge concentration.

Proof. The conclusion is immediate from the asymptotic assumption. Edge concentration is an integrated spatial property, whereas the linear coefficient only measures one asymptotic scale. $\square$

Proposition 21 (Stability under finite-depth local unitary processing). *Let V_L be a depth- d local unitary circuit with gate range r_0 , independent of L . A boundary projector supported on width ℓ is mapped to an operator supported within width $\ell + O(dr_0)$. Hence an extensive edge-concentration criterion is stable under such processing up to a subextensive change of boundary thickness.*

Proof. Each circuit layer enlarges the support of a boundary observable by at most the gate range. After d layers the enlargement is $O(dr_0)$. $\square$

XIV. DIAGNOSTIC SECTOR IV: DYNAMICAL CHANNEL SIGNATURES

For a localized input,

$$P_t(n|m) = \frac{|U_H(t)_{nm}|^2}{\sum_r |U_H(t)_{rm}|^2}. \quad (314)$$

Definition 39 (Localized reflection asymmetry). *For a site m sufficiently far from boundaries so that the reflection $n \mapsto 2m - n$ is defined on the relevant support, let*

$$(\mathcal{R}_m P)(n|m) = P(2m - n|m). \quad (315)$$

Define

$$D(t; m) = \frac{1}{2} \|P_t(\cdot|m) - \mathcal{R}_m P_t(\cdot|m)\|_1. \quad (316)$$

Definition 40 (Dynamical witnesses). *Define*

$$R_{\text{dyn}}^{\max}(H) = \sup_{t>0} \sup_m D(t; m), \quad (317)$$

and

$$R_{\text{dyn}}^{\infty}(H) = \limsup_{t \rightarrow \infty} \sup_m D(t; m). \quad (318)$$

These are operational witnesses, not QRT monotones.

Proposition 22 (Dynamical gauge embedding). *Let*

$$H_g = S_g H_0 S_g^{-1}, \quad S_g |n\rangle = e^{-gn} |n\rangle, \quad H_0^T = H_0. \quad (319)$$

Then

$$U_g(t) = S_g U_0(t) S_g^{-1}, \quad (320)$$

and whenever the denominator is nonzero,

$$\frac{U_g(t)_{nm}}{U_g(t)_{mn}} = e^{-2g(n-m)}. \quad (321)$$

Thus

$$a_{\text{dyn}}(t; r) = \frac{1}{2r} \ln \left| \frac{U_g(t)_{r0}}{U_g(t)_{0r}} \right| = -g \quad (322)$$

whenever the ratio is defined.

Proof. The similarity relation and $U_0(t)^T = U_0(t)$ imply

$$\frac{U_g(t)_{nm}}{U_g(t)_{mn}} = \frac{e^{-g(n-m)} U_0(t)_{nm}}{e^{g(n-m)} U_0(t)_{mn}} = e^{-2g(n-m)}. \quad (323)$$

□

Proposition 23 (Reciprocal stochastic post-processing contracts dynamical asymmetry). *Let K_t be a stochastic kernel commuting with reflection around m . Then*

$$D(K_t P_t) \leq D(P_t). \quad (324)$$

Proof. Use ℓ^1 contractivity:

$$D(K_t P_t) = \frac{1}{2} \|K_t(P_t - \mathcal{R}_m P_t)\|_1 \leq \frac{1}{2} \|P_t - \mathcal{R}_m P_t\|_1. \quad (325)$$

□

Proposition 24 (Operational lower bound from transition amplitudes). *For the Julia-dilated channel,*

$$A_\circ(\mathcal{N}_{H,t}) \geq \frac{1}{2} \max_m \sum_n ||K_{H,t}(n, m)|^2 - |K_{H,t}(m, n)|^2|. \quad (326)$$

Proof. Prepare the physical input $|m\rangle\langle m|$ and perform the projective measurement whose outcomes include the physical-sector site states. Under $\mathcal{N}_{H,t}$ the corresponding physical-site probabilities are $|K_{H,t}(n, m)|^2$, while under $\mathcal{N}_{H,t}^{\text{rev}}$ they are $|K_{H,t}(m, n)|^2$. Data processing for trace distance and the defining optimization of the diamond norm yield the stated bound. □

Proposition 25 (Finite-range weighted propagation bound). *Let H be bounded and finite range. Let the position operator satisfy*

$$X|n, \alpha\rangle = n|n, \alpha\rangle. \quad (327)$$

For $\mu > 0$, define

$$J_{\sigma\mu} = \|e^{\sigma\mu X} H e^{-\sigma\mu X}\|_\infty, \quad \sigma = \pm 1. \quad (328)$$

Then

$$|U_H(t)_{nm}| \leq \begin{cases} \exp[J_{+\mu} t - \mu(n-m)], & n \geq m, \\ \exp[J_{-\mu} t - \mu(m-n)], & n < m. \end{cases} \quad (329)$$

Hence

$$|U_H(t)_{nm}| \leq \exp[-\mu|n-m| + J_\mu^* t], \quad J_\mu^* = \max\{J_{+\mu}, J_{-\mu}\}. \quad (330)$$

Proof. Conjugating the exponential by $e^{\sigma\mu X}$ and taking operator norms gives

$$\|e^{\sigma\mu X} U_H(t) e^{-\sigma\mu X}\|_\infty \leq e^{J_{\sigma\mu} t}. \quad (331)$$

Taking matrix elements yields the result. □

XV. EXACT HATANO–NELSON/RESOURCE EQUIVALENCE

Proposition 26 (Transpose-reciprocity completeness of the uniform Hatano–Nelson family). *Consider the thermodynamic uniform Hatano–Nelson family*

$$H_{g,L} = m\mathbb{1} + J e^g T_+ + J e^{-g} T_-, \quad T_- = T_+^T, \quad J > 0. \quad (332)$$

Then

$$H_{g,L}^T = H_{g,L} \iff g = 0. \quad (333)$$

Moreover, for $g \neq 0$,

$$\rho_{\text{GBZ}} = \frac{|g|}{a} > 0, \quad (334)$$

and the OBC right eigenvectors obey

$$\psi_{j,g}(n) \propto e^{-gn} \sin\left(\frac{\pi j n}{L+1}\right). \quad (335)$$

Thus conventional OBC skin accumulation occurs for every fixed $g \neq 0$. Consequently, the uniform Hatano–Nelson family is transpose-reciprocity complete with respect to conventional NHSE:

$$g \neq 0 \iff H_{g,L}^T \neq H_{g,L} \iff \rho_{\text{GBZ}} > 0 \iff \text{conventional OBC NHSE.} \quad (336)$$

Proof. The transpose exchanges the two nonreciprocal hopping amplitudes:

$$H_{g,L}^T = m\mathbb{1} + Je^{-g}T_+ + Je^gT_-. \quad (337)$$

Therefore

$$H_{g,L}^T = H_{g,L} \iff e^g = e^{-g} \iff g = 0. \quad (338)$$

The GBZ result follows from Proposition 15. For open boundaries,

$$H_{g,L} = S_g H_{0,L} S_g^{-1}, \quad S_g |n\rangle = e^{-gn} |n\rangle, \quad (339)$$

so right eigenvectors are obtained from reciprocal standing-wave eigenvectors by the exponential gauge factor e^{-gn} . Hence $g > 0$ produces left-edge accumulation, $g < 0$ produces right-edge accumulation, and $g = 0$ gives the reciprocal extended limit. This establishes the equivalence. $\square$

Corollary 8 (Exact trajectory-level NHSE/resource equivalence for uniform Hatano–Nelson). *Assume the operational time domain is $(0, \infty)$ and use the tight normalization of Definition 22. Then*

$$A_\diamond^{\text{sup}}(H_g) > 0 \iff g \neq 0 \iff \rho_{\text{GBZ}} > 0 \iff \text{conventional OBC NHSE.} \quad (340)$$

Moreover, for every $g \neq 0$,

$$\liminf_{t \rightarrow 0^+} \frac{A_{\diamond, \infty}(t)}{t^2} \geq \frac{2J^2 \sinh(2|g|)}{1 + \varepsilon_0} > 0. \quad (341)$$

Proof. The first line follows by combining Proposition 26 with Theorem 7. The thermodynamic bound is Proposition 17 with $\kappa_0 = 1 + \varepsilon_0$. $\square$

Remark 22 (Scope of the exact Hatano–Nelson theorem). *Theorem 8 is an exact solved subtheory. Its biconditional NHSE statement relies on the fact that, within the uniform Hatano–Nelson family, conventional OBC skin accumulation is equivalent to transpose-reciprocity breaking. The theorem is therefore not a claim that $A_\diamond^{\text{sup}} > 0$ is a universal detector of every possible NHSE mechanism.*

Proposition 27 (Multichannel cancellation does not invalidate the scope theorem). *For the direct-sum family*

$$H_{\oplus,g}(k) = H_g(k) \oplus H_{-g}(k), \quad g \neq 0, \quad (342)$$

the determinant winding may vanish,

$$W_{\oplus,g}(E) = 0, \quad (343)$$

while

$$H_{\oplus,g}^T \neq H_{\oplus,g} \quad (344)$$

and therefore

$$A_\diamond^{\text{sup}}(H_{\oplus,g}) > 0. \quad (345)$$

Thus the channel resource detects transpose-reciprocity breaking even when a single determinant-winding scalar cancels between channels.

Proof. The opposite Hatano–Nelson sectors contribute opposite determinant windings, so the total winding cancels. However,

$$H_{\oplus,g}^T = H_{-g} \oplus H_g \neq H_g \oplus H_{-g} = H_{\oplus,g} \quad (346)$$

for $g \neq 0$ in the fixed channel basis. Theorem 7 therefore yields positive trajectory-derived resource. $\square$

XVI. EXTENSIVE CELLULAR RESOURCE

The following extensive construction is deliberately formulated at the level of independent channel subsystems. A direct-sum single-particle Hamiltonian does not in general produce a tensor product of Julia channels under a single global normalization, so no such identification is made.

Definition 41 (Cellular channel array). *A cellular array consists of independent cell subsystems with primitive channel objects $\mathcal{N}_{c_1}, \dots, \mathcal{N}_{c_n}$ and composite channel*

$$\mathcal{N} = \bigotimes_{k=1}^n \mathcal{N}_{c_k}. \quad (347)$$

If each cell object is generated by a Hamiltonian H_{c_k} , its own normalization gauge and canonical Julia completion are used before tensor-product assembly. This tensor-product assembly is an operational cellular construction and is not identified with the Julia channel obtained by applying one global normalization to a single-particle direct-sum Hamiltonian.

Definition 42 (Spatially local free superchannel). *A free superchannel on the cellular array is spatially local if*

$$\Theta = \bigotimes_{k=1}^n \Theta_k, \quad \Theta_k \in \mathfrak{D}_L. \quad (348)$$

Denote this subclass by $\mathfrak{D}_{\text{loc}}$.

Remark 23. *This is a strict cellwise locality class. More general finite-depth, bounded-range superchannels would require an explicit rule for how the spatial partition is transformed and how inter-cell memories are treated.*

Definition 43 (Block-additive fixed-time resource). *For*

$$\mathcal{N} = \bigotimes_{k=1}^n \mathcal{N}_k, \quad (349)$$

define

$$E_{\diamond}(\mathcal{N}) = \sum_{k=1}^n A_{\diamond}(\mathcal{N}_k), \quad (350)$$

and

$$E_{\text{rob}}(\mathcal{N}) = \sum_{k=1}^n R_{\text{rob}}(\mathcal{N}_k). \quad (351)$$

Theorem 11 (Extensivity and local monotonicity). *For every spatially local free superchannel,*

$$E_{\diamond}(\Theta(\mathcal{N})) \leq E_{\diamond}(\mathcal{N}), \quad E_{\text{rob}}(\Theta(\mathcal{N})) \leq E_{\text{rob}}(\mathcal{N}). \quad (352)$$

For a homogeneous array of n identical cells with $A_{\diamond}(\mathcal{N}_c) = a > 0$,

$$E_{\diamond}(\mathcal{N}) = na. \quad (353)$$

Proof. Apply the fixed-time monotonicity theorem to each cell and sum. $\square$

Definition 44 (Time-optimized block-additive resource). *For a collection of independent cell Hamiltonians $\{H_{c_k}\}_{k=1}^n$, define*

$$E_{\diamond}^{\text{sup}} = \sum_{k=1}^n A_{\diamond}^{\text{sup}}(H_{c_k}), \quad E_{\text{rob}}^{\text{sup}} = \sum_{k=1}^n R_{\text{rob}}^{\text{sup}}(H_{c_k}). \quad (354)$$

Theorem 12 (Trajectory-derived extensivity and rescaling invariance). *Under cellwise free trajectory transformations,*

$$E_{\diamond}^{\text{sup}}(\mathfrak{f}\mathfrak{N}) \leq E_{\diamond}^{\text{sup}}(\mathfrak{N}). \quad (355)$$

and similarly for $E_{\text{rob}}^{\text{sup}}$. Moreover $E_{\diamond}^{\text{sup}}$ is additive under independent cell concatenation and invariant under a common positive energy rescaling of all cell Hamiltonians.

Proof. Apply trajectory-level monotonicity and tight-gauge energy-rescaling invariance to every cell, then sum. $\square$

Theorem 13 (Complete witness on the Hatano–Nelson gauge sector). *Let every cell object be drawn from a Hatano–Nelson gauge orbit. Then*

$$E_{\diamond}^{\text{sup}} > 0 \iff \text{at least one cell has } A_{\diamond}^{\text{sup}} > 0. \quad (356)$$

For a cell that is itself a thermodynamic uniform Hatano–Nelson family, Theorem 8 identifies this positivity with conventional OBC skin accumulation in that cell family.

Proof. The sum is nonnegative, so it is positive exactly when at least one summand is positive. The Hatano–Nelson identification follows from Theorem 8. $\square$

Remark 24 (Scope of the extensive extension). *The quantity $E_{\diamond}^{\text{sup}}$ is an operationally extensive resource under explicitly cellwise local free operations. It should not be confused with an intrinsic tensor-power rate of the global single-particle QRT, which collapses by Proposition 5. Nor does a finite cell by itself constitute a thermodynamic NHSE system; the conventional NHSE interpretation applies when that factor is a thermodynamic cell family.*

XVII. TOWARD SUFFICIENCY: AN EXPLICIT OBSTRUCTION AND A CONJECTURAL ATTENUATION PROTOCOL

Corollary 3 and Theorem 6 together give a necessary condition and a no-go result on the calibration orbit, but no sufficiency statement: we do not know whether $\mathcal{N}_{g,\tau_*} \succeq_{\mathfrak{D}} \mathcal{N}_{g',\tau_*}$ actually holds for $|g| > |g'|$ in the full ancilla-assisted class. This section isolates precisely what is already known, exhibits an explicit obstruction that rules out the most naive candidate protocol, and states the remaining gap as two concrete conjectures.

A. What is already solved: resource-value sufficiency

Proposition 13 already shows that *some* channel realizing any prescribed smaller resource value is always reachable: for $\mathcal{N} = \mathcal{N}_{g,\tau_*}$ and any $a' \in [0, A_{\diamond}(\mathcal{N}_{g,\tau_*})]$, choosing $p = a'/A_{\diamond}(\mathcal{N}_{g,\tau_*})$ and $\mathcal{R}_0 \in \mathfrak{F}_L$ arbitrary gives a free superchannel Θ_p with $A_{\diamond}(\Theta_p(\mathcal{N}_{g,\tau_*})) = a'$. We record this explicitly because it is easy to conflate with the open problem below, and it is not: $\Theta_p(\mathcal{N}_{g,\tau_*})$ is, for generic $\mathcal{R}_0$, not itself a Julia-dilated Hatano–Nelson calibration channel at all, let alone $\mathcal{N}_{g',\tau_*}$ for the specific g' with $A_{\diamond}(\mathcal{N}_{g',\tau_*}) = a'$. Matching the scalar monotone is therefore a strictly weaker goal than the one that matters physically: converting one calibration point into another.

Definition 45 (Orbit-restricted convertibility). Fix g and g' with $0 \leq |g'| \leq |g| \leq g_{\max}$ on the common-normalization domain of Theorem 5. We say the calibration is orbit-sufficient at (g, g') if there exists a finite-size free superchannel $\Theta \in \mathfrak{D}_L$ (ancilla-assisted, of any finite ancilla dimension) such that

$$\boxed{\Theta(\mathcal{N}_{g, \tau_*}) = \mathcal{N}_{g', \tau_*}} \quad (357)$$

exactly, as opposed to merely $A_\diamond(\Theta(\mathcal{N}_{g, \tau_*})) = A_\diamond(\mathcal{N}_{g', \tau_*})$.

B. An explicit obstruction: the naive gauge-similarity map is not physical

The gauge family itself is generated by the non-unitary similarity $H_g = S_g H_0 S_g^{-1}$, $S_g = \text{diag}(1, e^{-g})$, and the same similarity acts correctly at the level of the contraction K_g : a direct computation shows that conjugating K_g by S_δ shifts the gauge parameter exactly, $S_\delta K_g S_\delta^{-1} \propto K_{g+\delta}$ entrywise. It is therefore tempting to look for a free superchannel built directly from this similarity. The next proposition shows that the literal similarity map itself is not a quantum channel. This rules out that direct construction, but does not exclude more elaborate completely positive constructions that modify or embed the similarity before producing the desired channel transformation.

Proposition 28 (Non-physicality of scalar gauge conjugation). Let $S_\delta = \text{diag}(1, e^{-\delta})$ on the two-site space, $\delta \neq 0$ real, and define

$$S_\delta(X) = S_\delta X S_\delta^{-1}. \quad (358)$$

Then S_δ does not map density matrices to density matrices: there exists a rank-one density matrix ρ for which $S_\delta(\rho)$ is not Hermitian.

Proof. Take $\rho = |\psi\rangle\langle\psi|$ with $|\psi\rangle = (|1\rangle + |2\rangle)/\sqrt{2}$, so $\rho = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. With $S_\delta = \text{diag}(1, e^{-\delta})$, $S_\delta^{-1} = \text{diag}(1, e^\delta)$,

$$S_\delta(\rho) = \frac{1}{2} \begin{pmatrix} 1 & e^\delta \\ e^{-\delta} & 1 \end{pmatrix}. \quad (359)$$

This real matrix is symmetric, and hence Hermitian, if and only if $e^\delta = e^{-\delta}$, i.e. $\delta = 0$. For $\delta \neq 0$ it is not Hermitian and therefore not a density matrix. $\square$

Remark 25 (What the obstruction rules out, and what it does not). Proposition 28 rules out only the single most naive candidate: implementing the gauge shift by literal conjugation with the (non-unitary, non-isometric) matrix that algebraically defines the Hatano–Nelson family. It does not rule out the existence of a genuinely completely positive, trace-preserving realization of the same net effect on K_g . In particular, Theorem 6 constrains only the subclass $\mathfrak{D}_{\text{uc}}(L)$ of unitary-congruence superchannels Θ_V ; it

says nothing about the strictly larger class of non-unitary reciprocal sandwiching maps $\Theta_\Gamma(\mathcal{N}) = \Gamma^{\text{rev}} \circ \mathcal{N} \circ \Gamma$ of Proposition 12, which requires only $\Gamma \in \text{Obj}_L$ (no unitarity), nor about fully ancilla-assisted superchannels with a joint Stinespring dilation. The obstruction therefore sharpens, rather than closes, the search: any successful attenuation protocol must act on the channel through its Julia-dilated, genuinely CP structure – e.g. through the nonlinear dependence of the defect operators $D_K, D_{K^\dagger}$ on K – and not through a direct transplant of the algebraic gauge similarity used merely to define the Hamiltonian family.

C. Conjectures

Conjecture 1 (Orbit-restricted sufficiency). For every g, g' with $0 \leq |g'| \leq |g| \leq g_{\max}$ on the common-normalization domain, the calibration is orbit-sufficient at (g, g') in the sense of Definition 45, using a superchannel $\Theta \in \mathfrak{D}_L$ with an ancilla of dimension bounded uniformly in g, g' (e.g. $\dim E \leq 4$, matching the dimension of the Julia space itself).

A weaker, more tractable stepping stone concerns attenuation by an infinitesimal amount, where a perturbative construction might be found even if the global statement resists direct attack.

Conjecture 2 (Infinitesimal sufficiency). For every $g \in (0, g_{\max})$ there exists $\eta_0 > 0$ and a smooth family $\{\Theta_\eta\}_{\eta \in [0, \eta_0]} \subset \mathfrak{D}_L$, with $\Theta_0 = \text{id}$, such that

$$\Theta_\eta(\mathcal{N}_{g, \tau_*}) = \mathcal{N}_{g-\eta, \tau_*} \quad \text{for all } \eta \in [0, \eta_0]. \quad (360)$$

Conjecture 2 is the natural infinitesimal form of Conjecture 1, and proving it would amount to showing that the tangent direction $-\partial_g \mathcal{N}_{g, \tau_*}$ at g lies in the tangent cone, at $\mathcal{N}_{g, \tau_*}$, of the orbit $\{\Theta(\mathcal{N}_{g, \tau_*}) : \Theta \in \mathfrak{D}_L\}$ of free superchannels. We do not carry out this tangent-space analysis here; it requires characterizing the local structure of $\mathfrak{D}_L$ near the identity superchannel acting on a fixed channel, which is a nontrivial infinite-dimensional (in the ancilla dimension) feasibility question even in its linearized form.

Remark 26 (A concrete numerical test). At fixed finite ancilla dimension d_E , the orbit-conversion problem can in principle be posed as a semidefinite feasibility problem using the Choi operator of the complete superchannel, with its linear CPTP/causality constraints, together with the linear covariance constraint

$$\Theta(\mathcal{N}^{\text{rev}}) = [\Theta(\mathcal{N})]^{\text{rev}} \quad (361)$$

and the linear output constraint for a fixed input channel

$$\Theta(\mathcal{N}_{g, \tau_*}) = \mathcal{N}_{g', \tau_*}. \quad (362)$$

By contrast, parametrizing Γ_{pre} and Γ_{post} independently by Choi matrices and then composing them introduces

bilinear dependence on those two variables, so that parametrization is not itself a standard SDP representation of the general superchannel.

A negative result at bounded d_E would not disprove Conjecture 1, which allows arbitrary finite ancilla dimension. A positive feasibility result for a small explicitly defined subclass, such as the single-channel sandwich construction of Proposition 12, would constitute genuine partial progress. Such a computation is distinct from the direct channel-evaluation scripts reported in Sec. XX. We have not carried it out and do not use any anticipated numerical outcome in support of the present claims.

Remark 27 (Status of this section). We have not resolved the sufficiency direction of the conversion preorder on the calibration orbit. What is established here is: (i) the existing resource-value dilution (Proposition 13) already answers a strictly weaker question and should not be mistaken for orbit-restricted sufficiency; (ii) the most obvious candidate construction for the latter is provably not a valid quantum channel (Proposition 28); and (iii) the rigidity of Theorem 6 applies only to the unitary-congruence subclass and does not bear on the (much larger) space of candidate non-unitary or ancilla-assisted attenuators, so the sufficiency question remains genuinely open rather than closed by that theorem. This narrows the search considerably relative to the blank statement “the full ancilla-assisted preorder remains open” given in Sec. VI, without claiming a resolution.

VIII. BUFFERED LOCALITY OF COMPRESSED CHANNELS FOR GENUINELY COUPLED CHAINS

A genuinely coupled single-particle chain does not carry a canonical spatial tensor-product factorization: its Hilbert space is the direct sum $\mathcal{H}_L = \bigoplus_n \langle_n$. Therefore the expression $\rho_V \otimes \mathbb{1}_{\bar{V}}/d_{\bar{V}}$ and a spatial partial trace are not defined on the primary single-particle QRT object without an additional local tensor-product realization. The local reduction below is consequently stated as a separate conditional architecture. For the native single-particle setting, Proposition 24 is the unconditional local probe statement used in this paper.

Definition 46 (Tensor-product local realization). A channel family admits a tensor-product local realization on a window decomposition if its underlying input and output Hilbert spaces are explicitly identified as

$$\mathcal{K}_L = \mathcal{K}_V \otimes \mathcal{K}_{\bar{V}}. \quad (363)$$

The local reduced channel construction below is asserted only for such realizations. No canonical tensor-product factorization is inferred from the single-particle spatial direct sum.

Definition 47 (Reduced local channel). Let $\mathcal{N}_L$ be a CPTP bistochastic channel in a tensor-product local realization and fix

$$\sigma_{\bar{V}} = \frac{\mathbb{1}_{\bar{V}}}{d_{\bar{V}}}. \quad (364)$$

Define

$$\mathfrak{R}_V[\mathcal{N}_L](\rho_V) = \text{Tr}_{\bar{V}}[\mathcal{N}_L(\rho_V \otimes \sigma_{\bar{V}})]. \quad (365)$$

For every fixed state $\sigma_{\bar{V}}$ the map $\rho_V \mapsto \mathfrak{R}_V[\mathcal{N}_L](\rho_V)$ is linear and CPTP, being the composition of the CPTP maps $\rho_V \mapsto \rho_V \otimes \sigma_{\bar{V}}$, $\mathcal{N}_L$, and $\text{Tr}_{\bar{V}}$. For $\sigma_{\bar{V}} = \mathbb{1}_{\bar{V}}/d_{\bar{V}}$ and unital $\mathcal{N}_L$ it is also unital, hence bistochastic.

Proof. Linearity, complete positivity and trace preservation are inherited from the three factors. For unitality,

$$\mathfrak{R}_V[\mathcal{N}_L](\mathbb{1}_V) = \text{Tr}_{\bar{V}}\left[\mathcal{N}_L\left(\frac{\mathbb{1}_V \otimes \mathbb{1}_{\bar{V}}}{d_{\bar{V}}}\right)\right] \quad (366)$$

$$= \text{Tr}_{\bar{V}}\left[\frac{\mathbb{1}_V \otimes \mathbb{1}_{\bar{V}}}{d_{\bar{V}}}\right] = \mathbb{1}_V. \quad (367)$$

Trace preservation was noted above. $\square$

Remark 28 (Where the affine structure lives, and what is withdrawn). Definition 46 presumes a tensor-product local realization $\mathcal{K}_L = \mathcal{K}_V \otimes \mathcal{K}_{\bar{V}}$. A candidate is second quantization to the free-fermion Fock space $\mathcal{F} = \mathcal{F}_V \otimes \mathcal{F}_{\bar{V}}$ of the Julia-dilated single-particle unitary. For a genuine many-body realization of this kind, $\mathfrak{R}_V[\mathcal{N}_L]$ is, by Definition 47, an honest linear CPTP bistochastic map on density operators of $\mathcal{F}_V$. We have not constructed or analyzed this channel here; in particular we have not shown that its transpose equals the reduced channel of the second-quantized transposed Julia unitary, nor computed its diamond-norm asymmetry.

An earlier version of this section asserted that this reduced map is affine, $\Lambda_V(\rho_V) + \Xi_V$, in the density operator ρ_V . That statement is withdrawn: it was a type error. The affine structure occurs only in the single-particle correlation-matrix representation of Gaussian states, where $C(t) = K C(0) K^\dagger$ and, for $C(0) = C_V \oplus \mathbb{1}_{\bar{V}}/d_{\bar{V}}$,

$$C_V(t) = K_{VV} C_V K_{VV}^\dagger + \frac{1}{d_{\bar{V}}} K_{V\bar{V}} K_{V\bar{V}}^\dagger. \quad (368)$$

This is an affine map on Hermitian matrices, not a quantum channel on states, and $A_\diamond$ is not defined for it. The offset $\Xi_V := d_{\bar{V}}^{-1} K_{V\bar{V}} K_{V\bar{V}}^\dagger$ is thus a property of the covariance-matrix representation. It enters no monotonicity statement in this paper, and it depends explicitly on the complement dimension $d_{\bar{V}}$ through its prefactor (see Sec. XX).

Definition 48 (Buffered central window). Let V_j be a central interval of width ℓ , and let $W_j \supset V_j$ contain a buffer of thickness b on both sides. Assume the buffered windows

are mutually disjoint. Let H_{W_j} denote the Hamiltonian restricted to W_j with couplings crossing the boundary of W_j removed. The isolated buffered channel is constructed using the same normalization gauge as the full channel comparison on the chosen operational interval. In particular, the comparison is required to use the same scalar normalization $\kappa(t)$ for the full and restricted channels; an alternative normalization may be used only when its induced discrepancy is absorbed into explicit buffer error constants below.

For a tensor-product local realization, the genuine reduced channel is

$$\mathcal{M}_{V_j}^{(L)} := \Re_{V_j}[\mathcal{N}_{H_L, t}], \quad (369)$$

and similarly, for the isolated buffered chain,

$$\mathcal{M}_{V_j}^{(W)} := \text{Tr}_{W_j \setminus V_j} \left[\mathcal{N}_{H_{W_j}, t} \left(\rho \otimes \frac{\mathbb{1}_{W_j \setminus V_j}}{d_{W_j \setminus V_j}} \right) \right]. \quad (370)$$

These maps are honest linear CPTP bistochastic channels on the tensor-product local Hilbert space. We do not decompose them as $\Lambda + \Xi$.

For the separate single-particle covariance-matrix diagnostic, define

$$\Xi_{V_j}^{(L)} = \frac{1}{d_{V_j}} K_{V_j V_j}^{(L)} \left(K_{V_j V_j}^{(L)} \right)^\dagger, \quad (371)$$

and define $\Xi_{V_j}^{(W)}$ analogously with the complement inside the isolated window. These offsets belong only to the covariance-matrix representation discussed in Remark 28. They are not components of the reduced channel $\mathcal{M}_{V_j}^{(L)}$ or $\mathcal{M}_{V_j}^{(W)}$.

The symbol Λ_{V_j} is reserved below for the separately defined compressed-contraction Julia channel of Remark 29. No equality, decomposition, or approximation between Λ_{V_j} and the genuine reduced channel $\mathcal{M}_{V_j}$ is assumed.

Assumption 3 (Optional offset-buffer diagnostic). *Fix a compact operational time interval $\mathcal{T}_0$ and a class of finite-range, uniformly bounded chains with a common strict-contraction bound $\|K_{H_L, t}\|_\infty \leq q < 1$. If one wishes to formulate a separate locality hypothesis for the covariance-matrix offset, assume that there are constants $C_\Xi, \mu_\Xi > 0$, independent of bulk position, system size, and $t \in \mathcal{T}_0$, such that*

$$\left\| \Xi_{V_j}^{(L)} - \Xi_{V_j}^{(W)} \right\|_\infty \leq C_\Xi e^{-\mu_\Xi b}. \quad (372)$$

This assumption concerns only the covariance-matrix diagnostic Ξ_V . It is not used in any channel monotonicity statement or in Proposition 29. In particular, the numerical results of Sec. XX do not support an exponential estimate for the normalized offset in the form above.

Proposition 29 (Unconditional buffer estimate for the compressed-contraction channel). *Let H_L be bounded,*

$\|H_L\|_\infty \leq h$, and of range R . Let V be a window of width ℓ , $W \supset V$ a window with buffer $b > R$, and $H_W = P_W H_L P_W$ the restriction of H_L to W , extended by 0 outside W . Fix a compact operational-time interval $\mathcal{T}_0 \subset [0, \infty)$ and assume that the full and buffered propagators use the same scalar normalization $\kappa(t)$ with

$$\|U_L(t)\|_\infty, \|U_W(t)\|_\infty \leq q\sqrt{\kappa(t)}, \quad q < 1, \quad (373)$$

and with $\inf_{t \in \mathcal{T}_0} \kappa(t) > 0$. Then, for every fixed $\mu > 0$, there is a constant $C_U = C_U(\ell, R, M, h, \mu, \mathcal{T}_0, \kappa)$ independent of L and of the position of V such that

$$\left\| K_V^{(L)} - K_V^{(W)} \right\|_\infty \leq C_U e^{-\mu b}. \quad (374)$$

Consequently, with

$$C_\Lambda = 2 \left(1 + \frac{q}{\sqrt{1-q^2}} \right) C_U, \quad (375)$$

one has

$$\left\| \Lambda_V^{(L)} - \Lambda_V^{(W)} \right\|_\diamond \leq C_\Lambda e^{-\mu b}, \quad (376)$$

and, by transpose invariance of the operator norm,

$$\left\| (\Lambda_V^{(L)})^{\text{rev}} - (\Lambda_V^{(W)})^{\text{rev}} \right\|_\diamond \leq C_\Lambda e^{-\mu b}. \quad (377)$$

Proof. Write the finite-range Hamiltonian as a sum of spatial displacement bands,

$$H_L = \sum_{r=-R}^R H_L^{(r)}. \quad (378)$$

Each band can be obtained by Fourier averaging of the on-site phase rotations generated by the position operator X and therefore obeys

$$\left\| H_L^{(r)} \right\|_\infty \leq \|H_L\|_\infty \leq h. \quad (379)$$

Consequently, the weighted Hamiltonian norms of Proposition 25 satisfy

$$J_\mu^\star \leq \sum_{r=-R}^R e^{\mu|r|} \left\| H_L^{(r)} \right\|_\infty \leq (2R+1) h e^{\mu R} =: \hat{J}_\mu, \quad (380)$$

which is independent of L .

Let

$$\Delta = H_L - (H_W \oplus 0). \quad (381)$$

Since $H_W = P_W H_L P_W$,

$$\Delta P_W = (\mathbb{1} - P_W) H_L P_W, \quad \|\Delta P_W\|_\infty \leq h. \quad (382)$$

Moreover, ΔP_W has support only on the R cells outside each boundary of W . Let Q be the projector onto

those cells. Every matrix element connecting V to Q has separation at least $b - R$, so Proposition 25 gives, for $0 \leq s \leq t$,

$$\left| (P_V U_L(t-s)Q)_{\alpha\beta} \right| \leq e^{-\mu(b-R) + \hat{J}_\mu t}. \quad (383)$$

The matrix $P_V U_L(t-s)Q$ has dimensions $(\ell M) \times (2RM)$, and hence its Frobenius norm is bounded by

$$\|P_V U_L(t-s)Q\|_\infty \leq \|P_V U_L(t-s)Q\|_F \leq M\sqrt{2\ell R} e^{-\mu(b-R) + \hat{J}_\mu t}. \quad (384)$$

Duhamel's formula gives

$$P_V[U_L(t) - U_W(t)]P_V = -i \int_0^t P_V U_L(t-s) \Delta P_W U_W(s) P_V ds, \quad (385)$$

and $\|U_W(s)\|_\infty \leq e^{hs}$. Therefore

$$\|P_V[U_L(t) - U_W(t)]P_V\|_\infty \leq h t M\sqrt{2\ell R} e^{\mu R + (\hat{J}_\mu + h)t} e^{-\mu b}. \quad (386)$$

Dividing by $\sqrt{\kappa(t)}$ yields

$$\left\| K_V^{(L)} - K_V^{(W)} \right\|_\infty \leq C_U(\ell, R, M, h, \mu, \mathcal{T}_0, \kappa) e^{-\mu b}, \quad (387)$$

with a constant uniform on $\mathcal{T}_0$ because $\kappa(t)$ is bounded away from zero there.

By the Lipschitz estimate used in Proposition 9,

$$\left\| \tilde{U}_{K_1} - \tilde{U}_{K_2} \right\|_\infty \leq \left(1 + \frac{q}{\sqrt{1-q^2}} \right) \|K_1 - K_2\|_\infty. \quad (388)$$

The same estimate holds after transpose because the operator norm is invariant under matrix transpose. Using

$$\|\mathcal{U}_1 - \mathcal{U}_2\|_\diamond \leq 2 \|U_1 - U_2\|_\infty \quad (389)$$

for unitary channels and the triangle inequality in the definition of $A_\diamond$, we obtain

$$\left| A_\diamond(\Lambda_V^{(L)}) - A_\diamond(\Lambda_V^{(W)}) \right| \leq C(q) C_U e^{-\mu b}. \quad (390)$$

The rev-estimate follows by applying the same argument to the transposed contractions $(K^{(L)})^T$ and $(K^{(W)})^T$. $\square$

Remark 29 (Meaning of Λ_V in this section: compressed-contraction channel). *From here on, and overriding any earlier phrase such as “linear (vacuum-background) part”, Λ_V denotes the following well-typed object. For a strict contraction $K = K_{H,t}$ on $\mathcal{H}_L$ and a window V , let $K_V = P_V K P_V|_{\mathcal{H}_V}$ be its compression, so $\|K_V\|_\infty \leq \|K\|_\infty \leq q < 1$, and set*

$$\Lambda_V := \tilde{\mathcal{E}}_{K_V} \quad (\text{canonical Julia completion of } K_V \text{ on } \mathcal{H}_V \oplus \mathcal{H}_V^{\text{env}}). \quad (391)$$

Since $(K^T)_V = (K_V)^T$, Corollary 1 gives $\Lambda_V^{\text{rev}} = \tilde{\mathcal{E}}_{(K^T)_V}$, and $A_\diamond(\Lambda_V)$ is the ordinary channel asymmetry of a bistochastic unitary channel. The superscripts (L) and (W) refer to compression of K from the full chain and of K

from the isolated buffered chain H_W , with the same scalar κ .

Λ_V is a new channel on the window, associated with a truncation of the propagator. It is not shown here to be a piece, marginal, or bound for the reduced channel of a many-body tensor-product realization (Remark 28); that relation is open. The offset Ξ_V is a covariance-matrix diagnostic that appears in no $A_\diamond$ statement.

Corollary 9 (Buffered asymmetry bound for the compressed-contraction Julia channel). *Under the hypotheses of Proposition 29, for every $t \in \mathcal{T}_0$,*

$$\left| A_\diamond(\Lambda_{V_j}^{(L)}) - A_\diamond(\Lambda_{V_j}^{(W)}) \right| \leq C_\Lambda e^{-\mu b}. \quad (392)$$

Proof. By the triangle inequality,

$$|A_\diamond(\mathcal{A}) - A_\diamond(\mathcal{B})| \leq \frac{1}{2} (\|\mathcal{A} - \mathcal{B}\|_\diamond + \|\mathcal{A}^{\text{rev}} - \mathcal{B}^{\text{rev}}\|_\diamond). \quad (393)$$

Apply Proposition 29. $\square$

Remark 30 (Optional one-sided variant). *Because rev is a continuous involution on the finite-dimensional superoperator space, one may replace the second linear-part buffer estimate by a finite-dimensional norm bound*

$$\|\mathcal{A}^{\text{rev}} - \mathcal{B}^{\text{rev}}\|_\diamond \leq C_{\text{rev}}(d_V) \|\mathcal{A} - \mathcal{B}\|_\diamond \quad (394)$$

for fixed window dimension d_V . The two-sided formulation used in the main theorem is preferable because it is directly testable and avoids importing a non-intrinsic continuity constant.

Remark 31 (How the buffer analysis is to be used). *The operational content of the coupled-chain extension is separated into two distinct objects. The exact QRT monotonicity applies to the compressed-contraction Julia channel $\Lambda_V = \tilde{\mathcal{E}}_{P_V K P_V}$ defined in Remark 29. Its approximation by the isolated buffered-window channel is controlled by Proposition 29. Separately, the quantity Ξ_V belongs only to the single-particle covariance-matrix representation and is not a component of the reduced CPTP channel $\mathfrak{R}_V[\mathcal{N}_L]$. No relation between Λ_V and $\mathfrak{R}_V[\mathcal{N}_L]$ is used or claimed in the present theorem.*

Definition 49 (Local compressed-channel resource). *For disjoint windows $V_1, \dots, V_J$,*

$$E_{\diamond, \ell, b}^{\text{loc}}(L, t) = \sum_{j=1}^J A_\diamond(\Lambda_{V_j}^{(L)}). \quad (395)$$

Define the isolated-buffer benchmark by

$$E_{\diamond, \ell, b}^{\text{iso}}(L, t) = \sum_{j=1}^J A_\diamond(\Lambda_{V_j}^{(W)}). \quad (396)$$

For the separate covariance-matrix diagnostic, define the aggregate offset sizes

$$\Xi_{\ell,b}^{\text{loc}}(L,t) = \sum_{j=1}^J \left\| \Xi_{V_j}^{(L)} \right\|_{\infty}, \quad \Xi_{\ell,b}^{\text{iso}}(L,t) = \sum_{j=1}^J \left\| \Xi_{V_j}^{(W)} \right\|_{\infty}. \quad (397)$$

These quantities are not identified with the channel resource $E_{\diamond,\ell,b}^{\text{loc}}$.

Proposition 30 (Exact local monotonicity for compressed channels). *For each window V_j , let*

$$\Theta_j \in \mathfrak{D}(V_j) \quad (398)$$

denote a free superchannel defined by the same covariance-preserving construction as $\mathfrak{D}_L$, with the Hilbert-space dimension and reciprocity frame restricted to that window. Assume that the Θ_j act independently on the compressed-contraction Julia channels $\Lambda_{V_j}^{(L)}$. Then

$$\sum_{j=1}^J A_{\diamond}(\Theta_j(\Lambda_{V_j}^{(L)})) \leq E_{\diamond,\ell,b}^{\text{loc}}(L,t). \quad (399)$$

No robustness aggregate is introduced in this section.

Proof. Each $\Lambda_{V_j}^{(L)}$ is a bistochastic Julia-completed channel on the window, so Theorem 1 gives

$$A_{\diamond}(\Theta_j(\Lambda_{V_j}^{(L)})) \leq A_{\diamond}(\Lambda_{V_j}^{(L)}). \quad (400)$$

Summing over j yields the claim. $\square$

Proposition 31 (Exponential coupling control for compressed-contraction channels). *Under the hypotheses of Proposition 29 (in particular $b > R$),*

$$\left| E_{\diamond,\ell,b}^{\text{loc}}(L,t) - E_{\diamond,\ell,b}^{\text{iso}}(L,t) \right| \leq J C(q) C e^{-\mu b}. \quad (401)$$

If fixed-width windows tile the bulk so that $J = \Theta(L)$, the discrepancy divided by L is $O(e^{-\mu b})$.

Proof. Sum the single-window estimate of Proposition 29 over j and use the triangle inequality. If $J = \Theta(L)$, division by L leaves the exponential dependence on b up to a constant prefactor. $\square$

Proposition 32 (Bulk local resource density). *Assume a fixed-width bulk tiling for which the number of windows satisfies*

$$\frac{J_L}{L} \rightarrow \nu_{\ell} \quad (402)$$

as $L \rightarrow \infty$, with finite ν_{ℓ} . Define

$$a_{\diamond,\ell,b}^{\text{loc}}(t) = \limsup_{L \rightarrow \infty} \frac{1}{L} E_{\diamond,\ell,b}^{\text{loc}}(L,t) \quad (403)$$

and

$$a_{\diamond,\ell,b}^{\text{iso}}(t) = \limsup_{L \rightarrow \infty} \frac{1}{L} E_{\diamond,\ell,b}^{\text{iso}}(L,t). \quad (404)$$

Then both densities are finite for fixed ℓ, t , and the unconditional buffer estimate gives

$$\left| a_{\diamond,\ell,b}^{\text{loc}}(t) - a_{\diamond,\ell,b}^{\text{iso}}(t) \right| \leq \nu_{\ell} C(q) C e^{-\mu b}. \quad (405)$$

No corresponding statement is made for the covariance-matrix offset.

Proof. Since $0 \leq A_{\diamond} \leq 1$,

$$0 \leq \frac{1}{L} E_{\diamond,\ell,b}^{\text{loc}}(L,t) \leq \frac{J_L}{L}, \quad (406)$$

so the local density is finite under the assumed bounded tiling density. The same argument applies to the isolated-buffer benchmark. Proposition 31 gives

$$\left| E_{\diamond,\ell,b}^{\text{loc}}(L,t) - E_{\diamond,\ell,b}^{\text{iso}}(L,t) \right| \leq J_L C(q) C e^{-\mu b}. \quad (407)$$

Dividing by L and taking the limsup yields the stated bound. $\square$

Remark 32. *These statements provide a controlled buffer analysis for coupled chains only when an explicit spatial tensor-product realization is supplied for any claim involving a genuine reduced channel. For the native single-particle Julia channel, the unconditional local probe is Proposition 24. The buffered theorem proved here concerns only the compressed contraction $K_V = P_V K P_V$ and its Julia-completed channel Λ_V . The covariance-matrix offset Ξ_V is a separate representation-dependent diagnostic and is not part of the channel statement.*

XIX. SYMMETRY-EXTENDED ARCHITECTURE

Definition 50 (Covariance datum). *A covariance datum on Obj_L is a map $r : \text{Obj}_L \rightarrow \text{Obj}_L$ satisfying*

1. $r^2 = \text{id}$;
2. r is convex-linear;
3. $r(\text{Obj}_L) \subseteq \text{Obj}_L$;
4. r acts cellwise on tensor-product arrays.

Transpose, $r(\mathcal{N}) = \mathcal{N}^{\text{rev}}$, is an example. Given a finite family $\mathcal{G}$ of covariance data, a superchannel is called $\mathcal{G}$ -covariant if

$$\Theta \circ r = r \circ \Theta \quad \forall r \in \mathcal{G}. \quad (408)$$

Definition 51 (Symmetry-extended free set and asymmetry). *For a finite family $\mathcal{G}$ of covariance data,*

$$\mathfrak{F}_{\mathcal{G}} = \bigcap_{r \in \mathcal{G}} \text{Fix}(r), \quad (409)$$

and

$$A_{\mathcal{G}}(\mathcal{N}) = \frac{1}{2} \max_{r \in \mathcal{G}} \|\mathcal{N} - r(\mathcal{N})\|_{\diamond}. \quad (410)$$

Theorem 14 (Symmetry-extended monotonicity). *For every $\mathcal{G}$ -free superchannel,*

$$A_{\mathcal{G}}(\Theta(\mathcal{N})) \leq A_{\mathcal{G}}(\mathcal{N}). \quad (411)$$

and

$$A_{\mathcal{G}}(\mathcal{N}) = 0 \iff \mathcal{N} \in \mathfrak{F}_{\mathcal{G}}. \quad (412)$$

On a cellular array,

$$E_{\mathcal{G}}(\mathcal{N}) = \sum_k A_{\mathcal{G}}(\mathcal{N}_k) \quad (413)$$

is monotone under symmetry-covariant cellwise free superchannels.

Proof. Let r^* attain the maximum in $A_{\mathcal{G}}(\Theta(\mathcal{N}))$. Covariance gives $r^*\Theta = \Theta r^*$, so

$$A_{\mathcal{G}}(\Theta(\mathcal{N})) = \frac{1}{2} \|\Theta(\mathcal{N}) - \Theta(r^*(\mathcal{N}))\|_{\diamond} \quad (414)$$

$$\leq \frac{1}{2} \|\mathcal{N} - r^*(\mathcal{N})\|_{\diamond} \quad (415)$$

$$\leq A_{\mathcal{G}}(\mathcal{N}). \quad (416)$$

Faithfulness follows from the finite maximum. $\square$

Remark 33. *The present paper uses the single covariance datum $\mathcal{G} = \{\text{rev}\}$. A different NHSE sector can be incorporated only after its physical symmetry action has been explicitly defined and shown to satisfy the covariance axioms. In particular, no reciprocal non-Abelian $\mathbb{Z}_2$ skin resource or Lindbladian skin resource is claimed to have been constructed here.*

A. Postselection versus genuinely open quantum dynamics

The present CPTP embedding is a completion of a normalized non-Hermitian propagation operator. It is not a claim of equivalence with arbitrary gain/loss Lindblad dynamics. A genuinely open system resource theory would require an enlarged object class, for example CPTP dynamical maps or Lindbladian generators, with its own free operations. This distinction is substantive: recent work has shown that Lindbladian skin topology can retain information that is invisible in postselected non-Hermitian Hamiltonian descriptions [14].

XX. NUMERICAL CHECKS AND REPRODUCIBILITY STATUS

The two-site calibration is fully analytic and can be checked directly from the four-by-four Julia unitary. No numerical data are needed for the central ordering and rigidity theorems. In particular, for

$$W_g = \tilde{U}_{K_g}^\dagger \tilde{U}_{K_g}^T, \quad (417)$$

the eigenphases determine

$$A_{\diamond} = \sqrt{1 - c(g)^2}. \quad (418)$$

Beyond the analytic calibration, three numerical checks are reported. They should be understood as consistency checks of the analytic statements rather than as independent proofs.

Lemma 1 (quasi-locality). At $L = 60$ and fixed t , the propagator and normalized contraction give the same fitted rate, $\mu(U_H(t)) = \mu(K) \approx 2.5648$ over the short range before reaching double-precision noise. For the defect operators, the quoted fit on $r \in [2, 25]$ gives $\mu(D_K) = \mu(D_{K^\dagger}) \approx 0.6625$, while refitting over the full available range gives approximately 0.5938. The visible dependence on the fit window is therefore non-negligible, and the numerical data should not be used to claim a unique precision exponent for the defect operators. The robust numerical conclusion is qualitative: the defect operators are exponentially decaying over the tested range but display a slower and more fit-sensitive decay than K itself.

Proposition 16 (thermodynamic persistence). For the uniform Hatano–Nelson family at fixed $g \neq 0$ and the tight gauge of Remark 20, the concrete finite-size lower-bound expression was evaluated for $L = 4, 8, 12, 20, 30, 45, 60, 90, 120$. The reported values decrease from approximately 0.1376 at $L = 4$ to 0.122676 at $L = 120$. This numerical sequence is consistent with the analytic L -uniform short-time estimate and illustrates that the finite-size expression does not visibly degrade with system size; it is not itself a proof of the thermodynamic bound.

Compressed-contraction buffer test (Sec. XVIII). For $\ell = 4$, bulk $L = 60$, and a common normalization inferred from the bulk propagator norm, the script evaluates the asymmetry of the compressed-contraction Julia channel $\Lambda_V = \tilde{\mathcal{E}}_{K_V}$ for the full chain and for isolated buffered windows. This is a test of Proposition 29, not of a many-body reduced channel. The bulk value is $A_{\diamond}(\Lambda_V^{(L)}) = 0.528469$, and the isolated-window value agrees to the displayed precision from $b = 2$ onward. The reported exponential fits are strongly dependent on the fitting range: using $b = 1, 2, 3, 4$ gives $\mu \approx 6.1525$, whereas using $b = 1, \dots, 6$ gives $\mu \approx 4.2686$. Because only a few points remain above numerical precision, these values should not be interpreted as reliable estimates of the analytic decay rate.

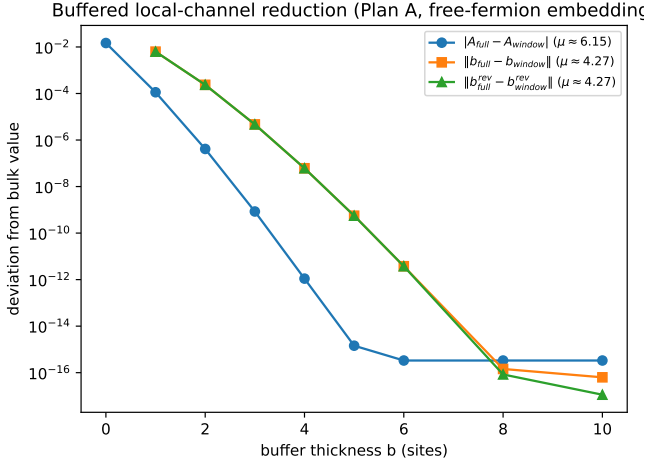


FIG. 3. Buffered convergence of the compressed-contraction Julia channel $\Lambda_V = \tilde{\mathcal{E}}_{P_V K P_V}$ for $\ell = 4$ and bulk $L = 60$. The asymmetry discrepancy $|A_\diamond(\Lambda_V^{(L)}) - A_\diamond(\Lambda_V^{(W)})|$ falls to the numerical floor rapidly; because the fitted exponent depends strongly on the fitting range, no precision value of μ_Λ is claimed. The covariance-matrix offset diagnostic behaves differently: over the tested range its discrepancy is consistent with approximately $1/b$ decay, attributable to the different maximally-mixed complement dimensions used in the full and isolated constructions. Accordingly, the offset is not presented as evidence for exponential buffer convergence. The figure concerns the compressed-contraction channel and the separate covariance diagnostic; it is not a test of a genuine many-body reduced channel.

We therefore use the numerical data only to demonstrate rapid convergence to the numerical floor, consistent with the analytic exponential buffer bound.

The offset diagnostic has a different behavior. The full-chain and window offsets use different maximally-mixed complement dimensions, $d_{\bar{V}} = 56$ and $d_{\bar{V}} = 2b$, respectively. The reported differences $\|\Xi^{(L)} - \Xi^{(W)}\| \approx 0.070, 0.035, 0.023, 0.016, 0.013, 0.010, 0.0068, 0.0049$ for $b = 1, 2, 3, 4, 5, 6, 8, 10$ satisfy

$$b \|\Xi^{(L)} - \Xi^{(W)}\| \in [0.049, 0.071],$$

which is consistent with approximately $1/b$ decay over the tested range. This behavior is dominated by the mismatched complement-dimension prefactor rather than by a demonstrated nonlocality of the propagator. Accordingly, no exponential fit is quoted for the normalized offset, and the earlier value $\mu_\Xi \approx 0.28$ is withdrawn as a locality exponent. Whether the un-normalized leakage $K_{V\bar{V}} K_{V\bar{V}}^\dagger$ has an independent exponential buffer limit remains untested and plays no role in the present channel results. The numerical study uses the single parameter point $(g, t, \ell, L) = (0.5, 0.35, 4, 60)$.

XXI. UNIFIED RESOURCE ARCHITECTURE

The logical structure is

fixed-time QRT	: $(\text{Obj}_L, \mathfrak{F}_L, \mathfrak{D}_L, \succeq_{\mathfrak{D}}, A_\diamond, R_\diamond)$
thermodynamic QRT layer	: $(\{\text{Obj}_L\}_{L \geq 1}, \mathfrak{D}_\infty, A_{\diamond, \infty}, R_\diamond)$
explicit rigid subclass	: $(\mathfrak{D}_{\text{uc}}, \chi_{\text{cs}})$
trajectory-derived extension	: $(A_\diamond^{\text{sup}}, R_\diamond^{\text{sup}})$
trajectory-level free maps	: $(\mathfrak{f} = \{\Theta_t\}_{t \in \mathcal{T}})$
cellular extensive extension	: $(\mathfrak{D}_{\text{loc}}, E_\diamond, E_{\text{rob}})$
trajectory-derived extensive extension	: $(E_\diamond^{\text{sup}}, E_{\text{rob}}^{\text{sup}})$
tensor-local coupled-chain extension	: $(E_{\diamond, \ell, b}^{\text{loc}}, a_{\diamond, \ell, b}^{\text{loc}}, \Xi_{\ell, b}^{\text{loc}})$
symmetry extension	: $(\mathcal{G}, \mathfrak{F}_{\mathcal{G}}, A_{\mathcal{G}})$
spectral diagnostics	: $(\beta, \rho_{\text{GBZ}}, W)$
disorder diagnostics	: $(\gamma, R_{\text{Lyp}}^{\text{skin}})$
spatial/dynamical diagnostics	: $(C_I^{L/R}, G_I; P_t, D_t)$

(419)

The exact logical implications are

$$H^T = H \implies \begin{cases} \mathcal{N}_{H,t}^{\text{rev}} = \mathcal{N}_{H,t}, \\ W_{\text{BZ}} = 0, \\ \beta \leftrightarrow \beta^{-1}, \\ \gamma_j = -\gamma_{2M+1-j}, \end{cases} \quad (420)$$

while

$$\mathcal{N} \succeq_{\mathfrak{D}} \mathcal{M} \implies A_\diamond(\mathcal{N}) \geq A_\diamond(\mathcal{M}), \quad R_{\text{rob}}(\mathcal{N}) \geq R_{\text{rob}}(\mathcal{M}). \quad (421)$$

For a cellular array,

$$\Theta \in \mathfrak{D}_{\text{loc}} \implies E_\diamond(\Theta(\mathcal{N})) \leq E_\diamond(\mathcal{N}), \quad (422)$$

and at the trajectory-derived level,

$$E_\diamond^{\text{sup}}(\mathfrak{f}\mathfrak{N}) \leq E_\diamond^{\text{sup}}(\mathfrak{N}). \quad (423)$$

For genuinely coupled chains, the correct local statement in the present paper is on the separately defined compressed-contraction Julia channels $\Lambda_{V_j}^{(L)} = \tilde{\mathcal{E}}_{P_{V_j} K P_{V_j}}$,

$$\sum_j A_\diamond(\Theta_j(\Lambda_{V_j}^{(L)})) \leq \sum_j A_\diamond(\Lambda_{V_j}^{(L)}), \quad (424)$$

together with the unconditional buffer estimate

$$\left| A_\diamond(\Lambda_{V_j}^{(L)}) - A_\diamond(\Lambda_{V_j}^{(W)}) \right| \leq \frac{C_\Lambda + C'_\Lambda}{2} e^{-\mu_\Lambda b}, \quad \left\| \Xi_{V_j}^{(L)} - \Xi_{V_j}^{(W)} \right\|_\infty \leq C \quad (425)$$

under Assumption 3. For the native single-particle direct-sum Hilbert space, the unconditional local probe is Proposition 24, not a partial-trace reduction.

On the uniform Hatano–Nelson orbit,

$$A_\diamond^{\text{sup}}(H_g) > 0 \iff g \neq 0 \iff \rho_{\text{GBZ}} > 0 \iff \text{conventional OBC} \quad (426)$$

with the NHSE statement understood for the thermodynamic uniform chain family. The full ancilla-assisted conversion problem remains distinct from this witness equivalence.

XXII. SCOPE AND LIMITATIONS

The primitive theory is finite-time and operational. At fixed finite L , its objects are the CPTP bistochastic channels on the doubled Hilbert space; the finite- L exponential locality notation is a compatibility structure, while non-trivial locality enters through L -uniform thermodynamic family conditions. The free channels are transpose reciprocal, and the free operations are covariance-constrained quantum superchannels that preserve the object class. We interpret this covariance condition as the axiomatic symmetry-free choice of the present QRT rather than as a complete microscopic model of engineering cost. Covariance is imposed on the stated object class; no stronger covariance on a larger space is silently assumed. The Julia dilation is a canonical representation choice for the normalized propagator, not a uniqueness theorem for physical dilations. The only no-ancilla rigidity statement proved here uses the explicitly defined unitary-congruence subclass $\mathfrak{D}_{\text{uc}}(L)$.

Several clarifications are central.

First, the finite-size quasi-locality definition is deliberately separated from the thermodynamic locality layer. The fixed- L object class is suitable for finite-dimensional QRT statements, while thermodynamic claims use sequences with L -uniform locality constants and superchannels with the locality certificate of Definition 14. Thus finite-dimensional global maps are not silently treated as thermodynamically local.

Second, channel transpose is a linear involution but an anti-homomorphism under composition:

$$(\mathcal{A} \circ \mathcal{B})^{\text{rev}} = \mathcal{B}^{\text{rev}} \circ \mathcal{A}^{\text{rev}}. \quad (427)$$

For this reason the reciprocity-covariance condition belongs to the complete superchannel. The main monotonicity theorems never use transpose as a diamond-norm isometry.

Third, the calibration family is ordered exactly by the monotone $A_{\diamond}$:

$$|g| \geq |g'| \iff A_{\diamond}(\mathcal{N}_{g,\tau_*}) \geq A_{\diamond}(\mathcal{N}_{g',\tau_*}). \quad (428)$$

This proves the necessary condition $|g| \geq |g'|$ for free conversion, and rules out deterministic gauge amplification. It does not establish deterministic attenuation in the full ancilla-assisted superchannel class. The ancilla-free unitary covariant subclass is rigid by the cosquare invariant. A full sufficiency theorem requires an explicit deterministic attenuation protocol.

Fourth, normalization dependence has been separated from the symmetry question. The condition $A_{\diamond} = 0$ is invariant under every finite strict scalar normalization, while the numerical value can change and can even be driven to zero by taking the normalization arbitrarily large. The quantitative normalization-equivalence theorem controls this dependence only inside a common strict-contraction ball. The trajectory-derived tight gauge removes the separate overall energy- rescaling ambiguity when the full positive-time domain is optimized.

Fifth, the exact generalization to NHSE is intentionally class based. Theorem 7 is a theorem about transpose-reciprocity breaking. A converse with conventional NHSE is available only on model classes in which conventional NHSE is itself equivalent to transpose-reciprocity breaking. The uniform Hatano–Nelson gauge orbit is the primary exact example. Reciprocal non-Abelian $\mathbb{Z}_2$ skin physics is explicitly outside this sector.

Sixth, the fixed-time global monotones are bounded by one, so their tensor-power rates vanish. Extensivity requires either the explicit tensor-product cellular construction with cellwise local operations or the local-channel construction for coupled systems, the latter only under an additional tensor-product local realization.

Seventh, a direct-sum single-particle Hilbert space does not possess a canonical spatial tensor product. Therefore the buffered reduction is conditional: it assumes an independently supplied tensor-product local realization if one wishes to speak of a many-body reduced channel; the buffer estimates proved here (Proposition 29) concern only the compressed contraction $P_V K P_V$ and its transpose. No partial trace over a single-particle spatial complement is taken.

Eighth, the coupled-chain buffer comparison is normalization-consistent by construction: the same scalar normalization is used for the full and isolated buffered channels, unless a separate normalization-discrepancy estimate is included in the explicit error constants. This prevents an unnoticed change of normalization gauge from being misidentified as a locality effect.

Ninth, the exact closed form of

$$A_{\diamond}^{\text{sup}}(H) = \sup_{t>0} A_{\diamond}(\mathcal{N}_{H,t}^{\text{tight}}) \quad (429)$$

beyond the two-site calibration and thermodynamic lower bound remains unresolved. The full ancilla-assisted covariant superchannel classification also remains open.

Tenth, the cellular extensive extension is an operational tensor-product construction. It is not asserted to equal the canonical Julia channel of a globally normalized single-particle direct-sum Hamiltonian. This distinction is essential because the scalar normalization and Julia defect operators generally do not factorize under a global normalization.

Eleventh, the reduced channel $\mathfrak{R}_V[\mathcal{N}_L]$ of Definition 47 is, for any tensor-product realization, a linear CPTP bistochastic map on density operators; the affine offset Ξ_V occurs only in the single-particle covariance-matrix representation of Gaussian states (Remark 28) and is not a property of that channel. The buffered analysis of Sec. XVIII is carried out on a different object: the Julia completion $\Lambda_V = \tilde{\mathcal{E}}_{K_V}$ of the compressed contraction $K_V = P_V K P_V$ (Remark 29), whose buffer convergence is an unconditional consequence of propagator quasi-locality (Proposition 29). We have not established any relation—equality, inequality, or approximation—between Λ_V and the reduced channel of a second-quantized (many-body) realization, nor computed the latter’s diamond-norm asymmetry. This

relation is the actual open problem of the coupled-chain extension. Numerically, only Λ_V was tested, at a single parameter point (Sec. XX); the covariance-matrix offset was found to decay as $1/b$ owing to a mismatched normalization and no exponential buffer convergence is claimed for it.

Finally, the present theory deliberately excludes genuinely Lindbladian resource objects. This is not a technicality: recent work distinguishes Lindbladian skin topology from postselected non-Hermitian Hamiltonian topology [14]. Likewise, reciprocal non-Abelian $\mathbb{Z}_2$ skin physics lies outside the transpose-reciprocity-breaking sector; a recent 2026 PRL provides an explicit reciprocal non-Abelian mechanism [13].

XXIII. CONCLUSION

We constructed a fixed-time quantum-channel resource theory for the transpose-reciprocity-breaking sector associated with conventional NHSE, with quasi-local CPTP bistochastic objects, reciprocal free channels, covariance-constrained free superchannels, and faithful monotones $A_\diamond$ and R_{rob} .

The Julia completion provides a canonical CPTP operational representation of normalized non-Hermitian propagation. The bistochastic object restriction closes the channel transpose operation, so the free set is mathematically well typed. The general Julia transpose identities are

$$D_K^T = D_{K^*}, \quad D_{K^\dagger}^T = D_{K^T}, \quad (430)$$

which give

$$\tilde{U}_K^T = \tilde{U}_{K^T}. \quad (431)$$

The canonical Julia choice is treated as a representation convention rather than as a uniqueness claim about all physical dilations.

The asymmetry has a direct operational meaning:

$$A_\diamond(\mathcal{N}) = 2p_{\text{succ}}^{\text{opt}}(\mathcal{N}, \mathcal{N}^{\text{rev}}) - 1. \quad (432)$$

Thus the resource is exactly twice the optimal one-shot advantage over random guessing when detecting whether a channel and its reciprocal transpose are being presented.

On the two-site Hatano–Nelson calibration, the exact channel asymmetry is strictly increasing in $|g|$ under a common strict normalization. Therefore the scalar parameter $|g|$ is exactly ordered by $A_\diamond$ on that calibration orbit, and every deterministic free conversion obeys the necessary condition $|g| \geq |g'|$. The stronger biconditional conversion law is not claimed because deterministic attenuation in the full ancilla-assisted class requires an explicit protocol. Within the explicitly defined ancilla-free unitary-congruence subclass, the phase-invariant cosquare scalar χ_{cs} makes $|g|$ rigid.

The normalization convention is treated as an operational gauge. Within the strict scalar class, the resource support is normalization independent, while quantitative values are controlled by an explicit continuity bound. Under the per-Hamiltonian tight gauge and full positive-time optimization,

$$A_\diamond^{\text{sup}}(\lambda H) = A_\diamond^{\text{sup}}(H). \quad (433)$$

The trajectory-derived extension is a pointwise derived construction: fixed-time monotonicity holds at each t , and taking the supremum yields monotonicity under families of pointwise free superchannels. This construction is not claimed to be a causal process-supermap resource theory.

The thermodynamic short-time theorem uses an explicitly L -uniform propagator remainder and an L -uniform transition-intensity estimate, numerically confirmed to converge cleanly with system size (Sec. XX). For the uniform Hatano–Nelson family this yields

$$\liminf_{t \rightarrow 0^+} \frac{A_{\diamond, \infty}(t)}{t^2} \geq \frac{2J^2 \sinh(2|g|)}{1 + \varepsilon_0} > 0 \quad (g \neq 0), \quad (434)$$

so the thermodynamic resource persistence is not a finite-size artifact at the level proved here.

The exact logical content of the NHSE correspondence is now separated into three layers. First, for finite-dimensional Hamiltonians with a time domain accumulating at zero,

$$A_{\diamond, \kappa}^{\text{sup}} > 0 \iff H^T \neq H. \quad (435)$$

Second, transpose-reciprocity-breaking conventional NHSE necessarily carries positive channel resource. Third, on transpose-reciprocity-complete model classes, conventional NHSE is exactly characterized by positivity of the trajectory-derived resource. The uniform Hatano–Nelson family is the principal exact example:

$$A_\diamond^{\text{sup}}(H_g) > 0 \iff g \neq 0 \iff \rho_{\text{GBZ}} > 0 \iff \text{conventional OBC} \quad (436)$$

The cellular extension gives an exactly additive extensive resource under an explicitly defined tensor-product laboratory architecture. For genuinely coupled systems, the present buffered theorem concerns only the compressed-contraction Julia channel $\Lambda_V = \tilde{\mathcal{E}}_{P_V K P_V}$. Its asymmetry obeys an unconditional exponential buffer estimate derived from propagator locality (Proposition 29). This result does not identify Λ_V with the reduced channel of a many-body tensor-product realization, nor does it provide such a reduced-channel approximation theorem. The covariance-matrix offset remains a separate representation-dependent diagnostic.

Spectral, disorder, spatial, and dynamical quantities remain diagnostics rather than axioms of the QRT. Clean reciprocity forces zero determinant winding and reciprocal roots; reciprocal disordered chains have Lyapunov pairing; and the $g \oplus (-g)$ construction demonstrates that scalar

winding can cancel while the channel resource remains positive.

The resulting architecture is therefore a genuine channel QRT for transpose-reciprocity breaking, supplemented by a trajectory-derived extension, an explicitly tensor-product extensive theory, and a compressed-contraction buffer analysis for coupled chains whose relation to a many-body reduced channel is explicitly left open. It is not a universal resource theory for the physical non-Hermitian skin effect, and we deliberately resist framing it as one: outside the uniform Hatano–Nelson calibration, the correspondence with conventional NHSE is one-directional (Corollary 4) or conditional on an explicitly stated completeness property of the model class (Definition 30). Three concrete gaps remain, in decreasing order of how fundamental we judge them to be. First, the numerical value of every monotone constructed here depends on an operator-chosen scalar normalization gauge (Proposition 10); only the resource *support* $A_\diamond = 0$ vs. > 0 is gauge invariant, and a gauge-free quantitative resource measure is not constructed. Second, only a necessary condition ($|g| \geq |g'|$) and a no-go rigidity result (Theorem 6)

are established for conversion on the calibration orbit; no sufficiency direction or explicit deterministic attenuation protocol is given for the full ancilla-assisted free-operation class. Third, the coupled-chain local analysis of Sec. XVIII is not closed: it is proved for the compressed-contraction channel $\Lambda_V = \tilde{\mathcal{E}}_{P_V K P_V}$, but no relation is established between Λ_V and the reduced channel of a second-quantized tensor-product realization of the chain, so a local resource statement for the many-body reduced channel is left as an explicit open problem. Extending the architecture to reciprocal non-Abelian $\mathbb{Z}_2$ skin physics or to genuinely Lindbladian skin topology would additionally require identifying the corresponding physical covariance data and enlarging the primitive object class.

ACKNOWLEDGMENTS

The authors thank colleagues for discussions on quantum resource theories, quantum channels and superchannels, non-Hermitian band theory, transfer matrices, and open-system non-Hermitian dynamics.

-
- [1] E. Chitambar and G. Gour, *Quantum resource theories*, Rev. Mod. Phys. **91**, 025001 (2019).
 - [2] Y. Liu and X. Yuan, *Operational resource theory of quantum channels*, Phys. Rev. Research **2**, 012035(R) (2020).
 - [3] G. Chiribella, G. M. D’Ariano, and G. M. Perinotti, *Transforming quantum operations: quantum supermaps*, Europhys. Lett. **83**, 30004 (2008).
 - [4] S. Yao and Z. Wang, *Edge states and topological invariants of non-Hermitian systems*, Phys. Rev. Lett. **121**, 086803 (2018).
 - [5] F. K. Kunst, E. Edvardsson, J. C. Budich, and E. J. Bergholtz, *Biorthogonal bulk-boundary correspondence in non-Hermitian systems*, Phys. Rev. Lett. **121**, 026808 (2018).
 - [6] N. Hatano and D. R. Nelson, *Localization transitions in non-Hermitian quantum mechanics*, Phys. Rev. Lett. **77**, 570 (1996).
 - [7] H. Ammari, S. Barandun, J. Cao, B. Davies, and E. O. Hiltunen, *Mathematical Foundations of the Non-Hermitian Skin Effect*, Arch. Ration. Mech. Anal. **248**, 33 (2024).
 - [8] K. Zhang, Z. Yang, and C. Fang, *Correspondence between winding numbers and skin modes in non-Hermitian systems*, Phys. Rev. Lett. **125**, 126402 (2020).
 - [9] N. Okuma, K. Kawabata, T. Shiozaki, and M. Sato, *Topological origin of non-Hermitian skin effects*, Phys. Rev. Lett. **124**, 086801 (2020).
 - [10] K. Kawabata, K. Shiozaki, M. Ueda, and M. Sato, *Symmetry and topology in non-Hermitian physics*, Phys. Rev. X **9**, 041015 (2019).
 - [11] A. Acín, *Statistical distinguishability between unitary operations*, Phys. Rev. Lett. **87**, 177901 (2001).
 - [12] S. Becker, N. Datta, L. Lami, and C. Rouzé, *Energy-constrained discrimination of unitaries, quantum speed limits, and a Gaussian Solovay–Kitaev theorem*, Phys. Rev. Lett. **126**, 190504 (2021).
 - [13] H. Tang, Y. Zhang, Z. Wang, L. Tang, D. Song, J. Xu, W. Zhang, H. Buljan, X. Zhang, and Z. Chen, *Non-Abelian Route to $\mathbb{Z}_2$ Non-Hermitian Skin Effects*, Phys. Rev. Lett. **137**, 126605 (2026).
 - [14] A. Chaduteau, D. K. K. Lee, and F. Schindler, *Lindbladian versus Postselected non-Hermitian Topology*, Phys. Rev. Lett. **136**, 016603 (2026).
 - [15] J. Sirker, *Pseudospectral phenomena and the origin of the non-Hermitian skin effect*, arXiv:2603.22643 (2026).